

Data Structure

Course No. 40989

Week #12

Red-Black Tree (Advanced)

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Red-Black Tree

What is a Red-Black Tree?

Definition

- A binary search tree in which every node is colored with either **Red** or **Black**
- It satisfies the red-black properties

(Recap) Binary Tree Definition

- A binary tree is a tree where every node has 0, 1, or 2 children.
- Each child node is labeled as either a left child or a right child.

(Recap) Binary Search Tree Definition

- A binary tree is a tree where every node has 0, 1, or 2 children.
- Binary Search Tree is a binary tree, in which for every node, the ***right*** subtree contains only nodes with values ***larger than*** the node's value, and ***left*** subtree contains only nodes with values ***smaller than*** the node's value.

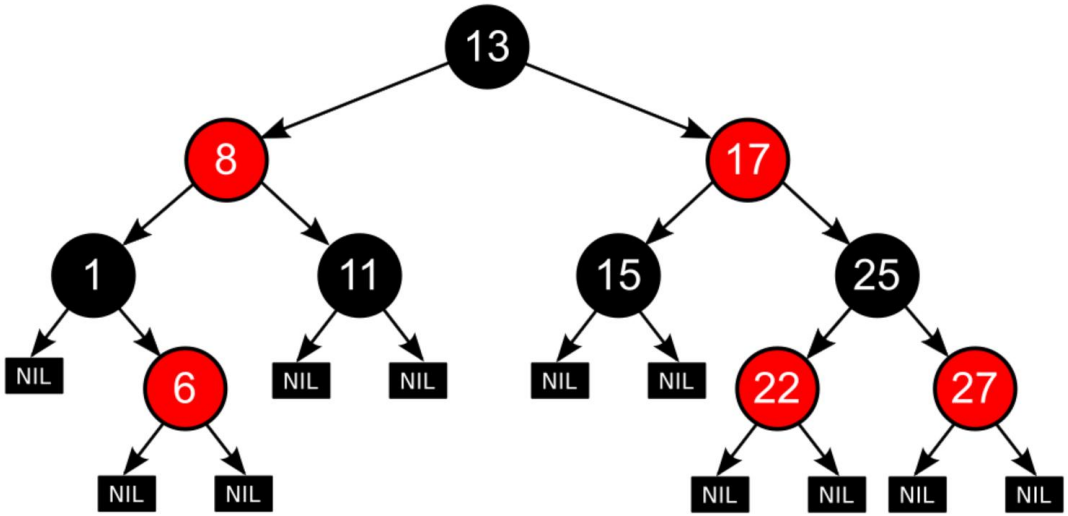
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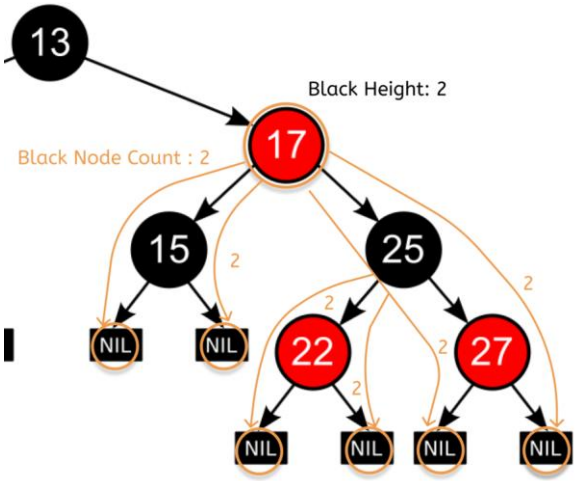
Red-Black Tree

What is a Red-Black Tree?

Red-Black Properties

- Every node is either **Red** or **Black**.
- The root is always **Black**.
- Every leaf (NULL=NIL) is **Black**.
- If a node is **Red**, both its children are **Black** (a **Black** node can have both **Red**/**Black** children).
- For each node, all paths from a node A to a NIL node B contain the same number of **Black** nodes.



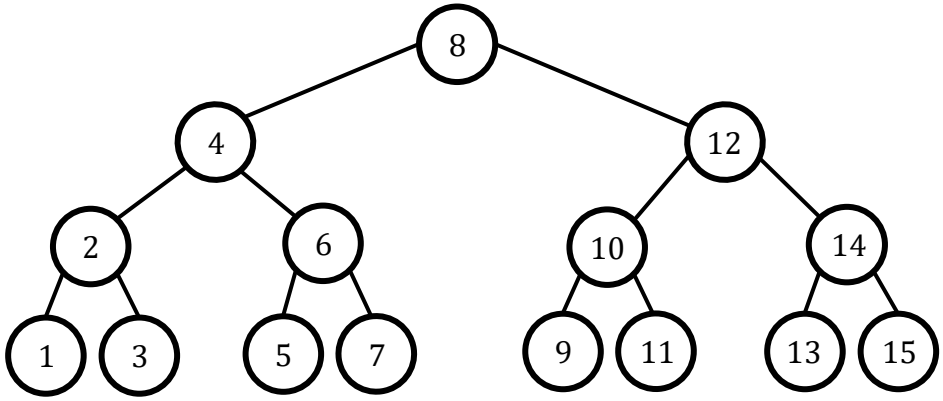


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Red-Black Tree

Exercise

Given a complete binary tree of height 3 with the key {1,2,3,4,…,15}, add the NIL leaves and color the nodes in three different ways such that the black heights of root '8' is 2, 3, and 4.

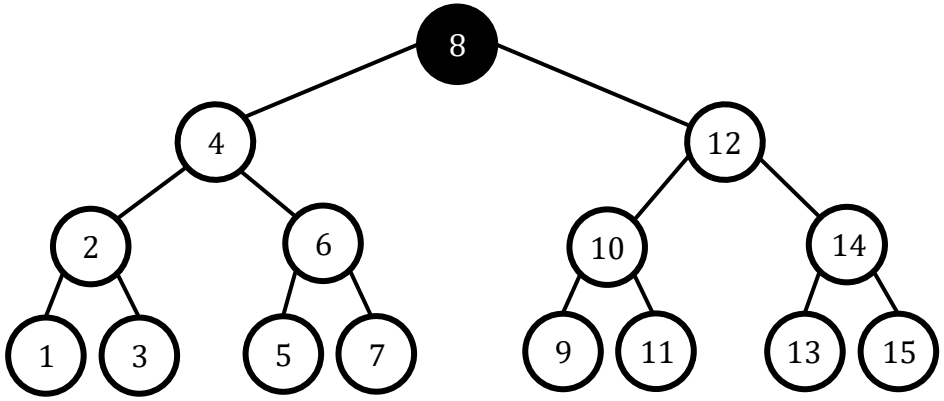


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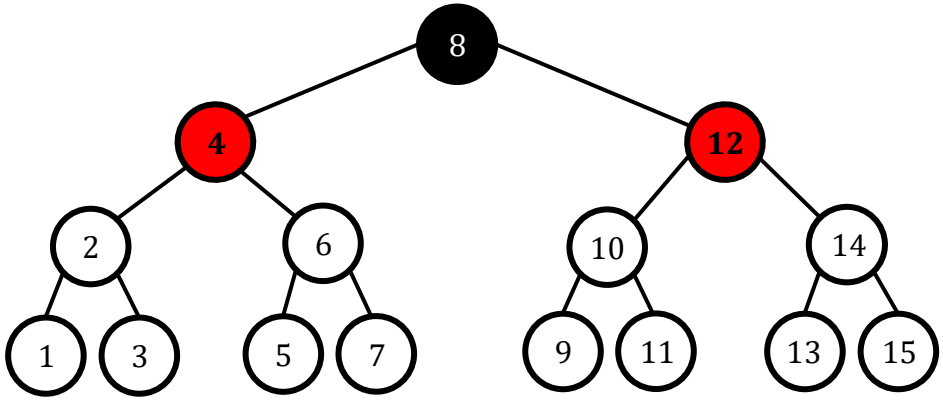


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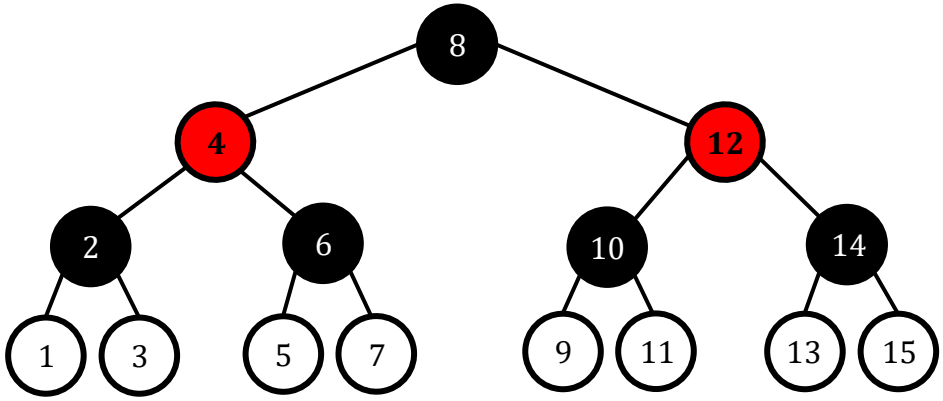


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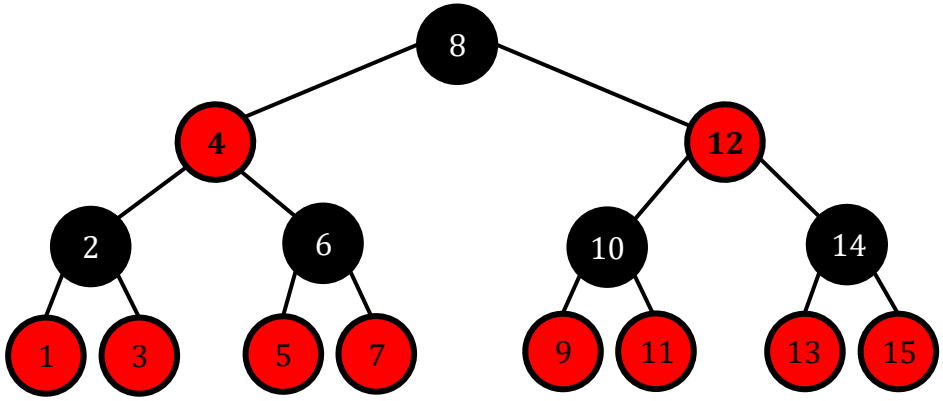


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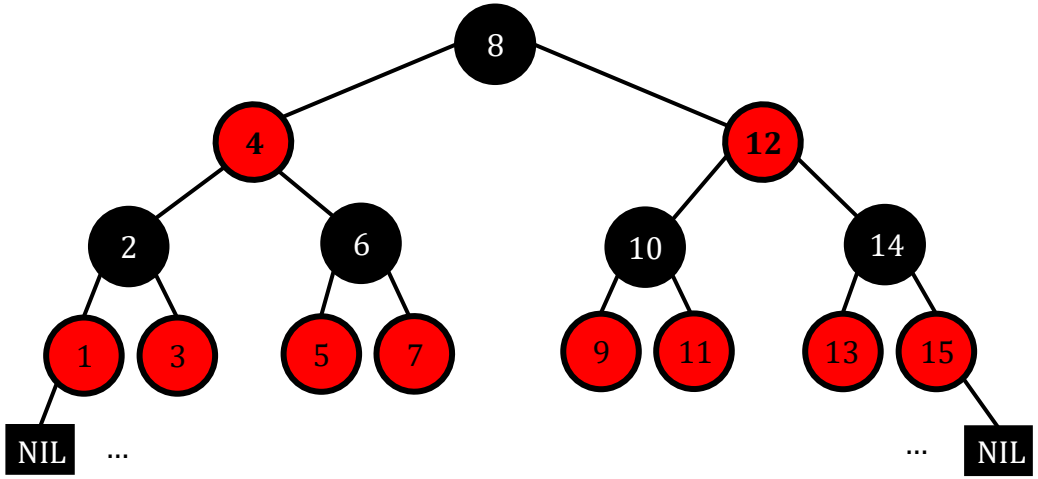


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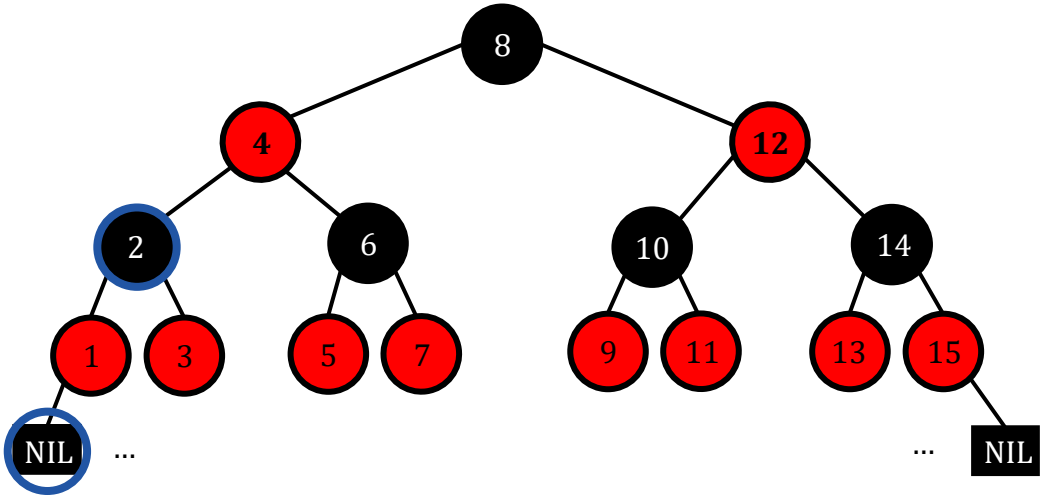


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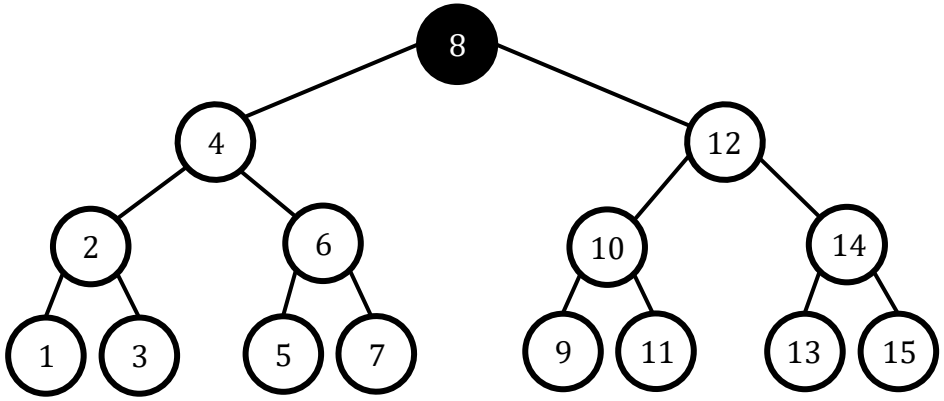


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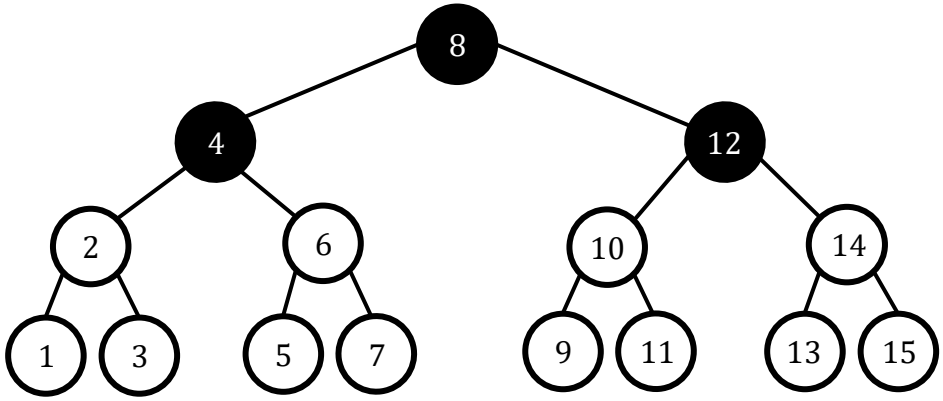


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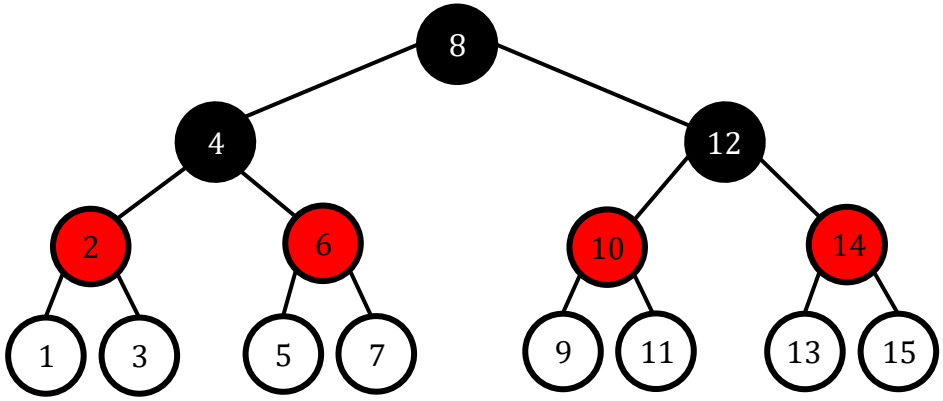


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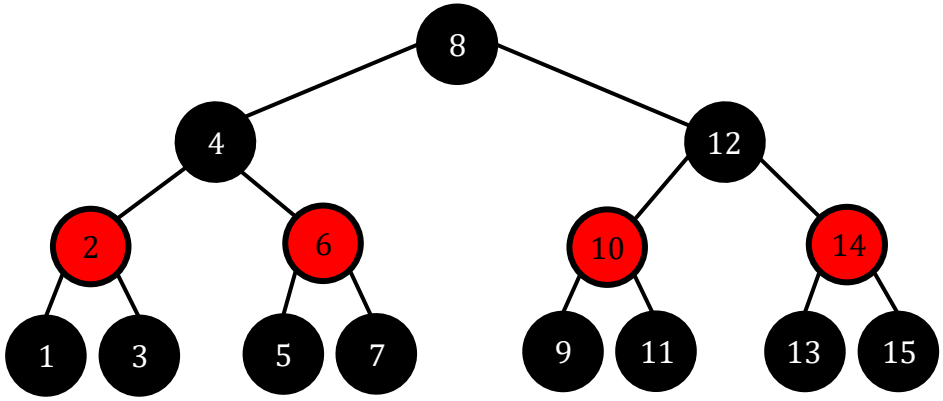


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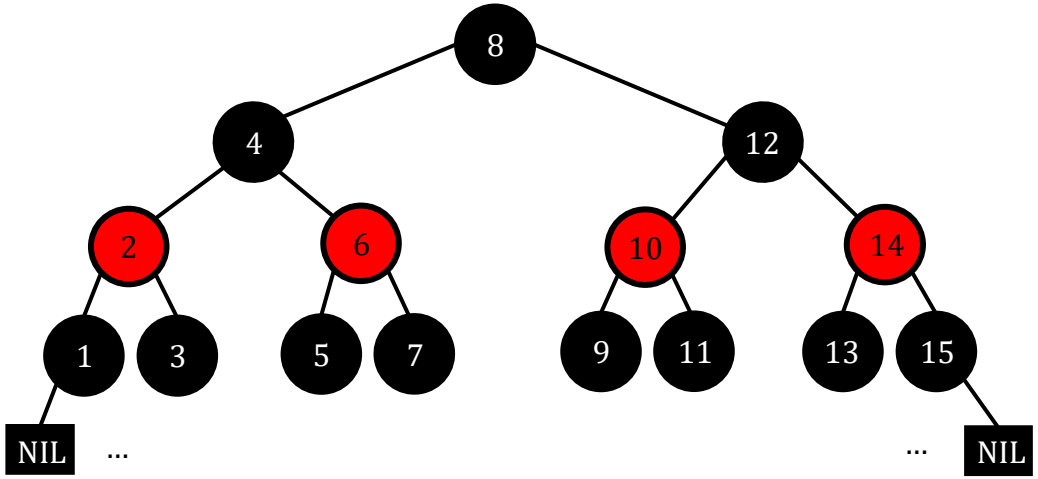


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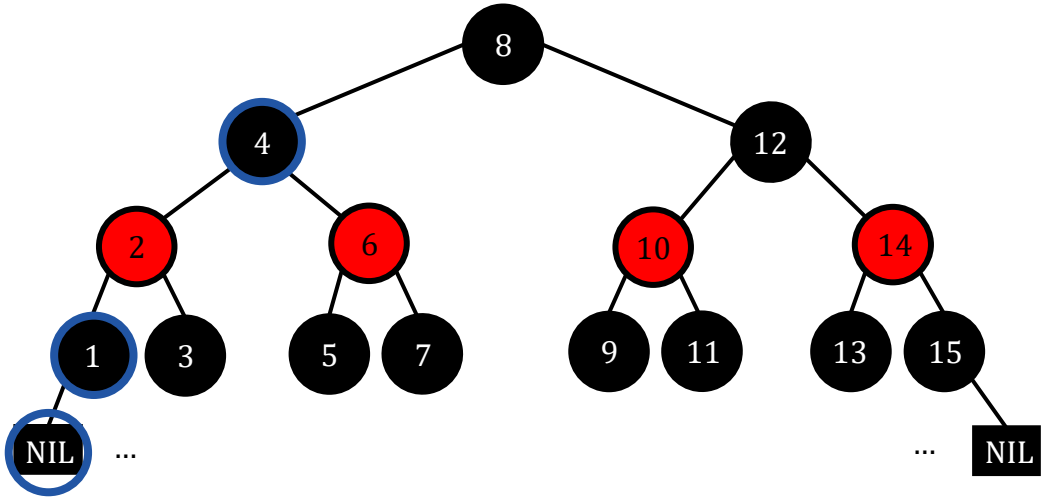


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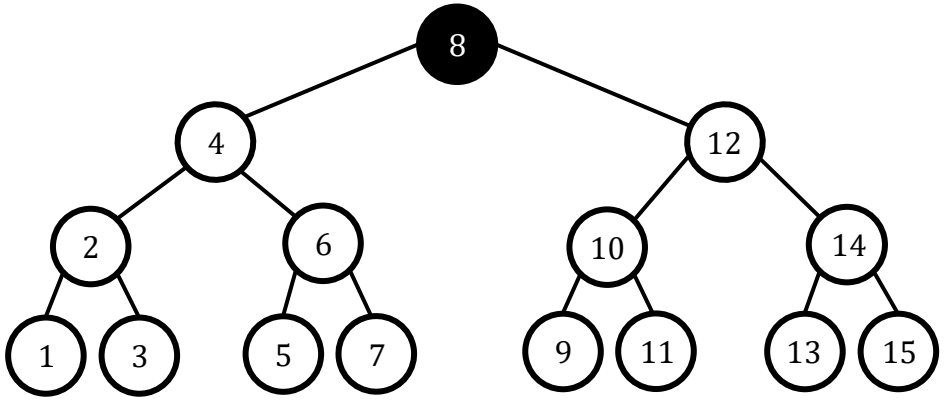


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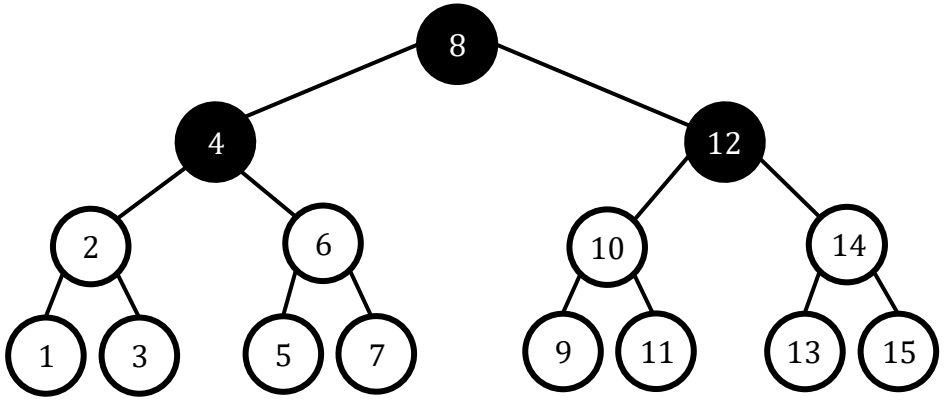


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Red-Black Tree

Exercise

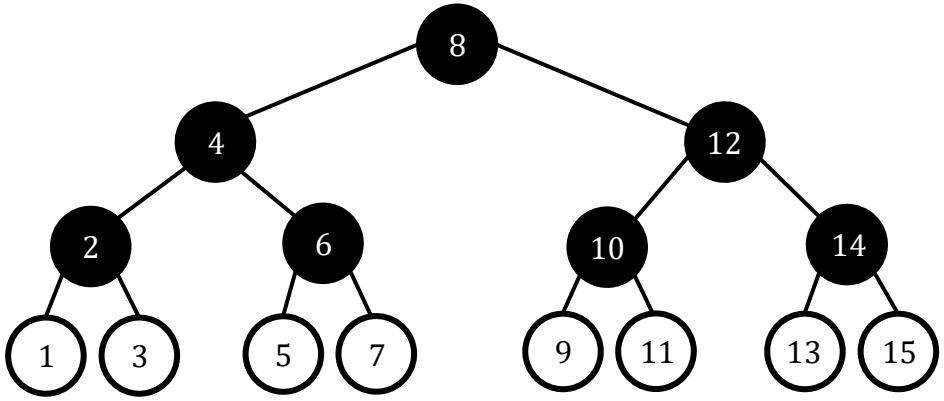
Given a complete binary tree of height 3 with the key {1,2,3,4,…,15}, add the NIL leaves and color the nodes in three different ways such that the black heights of root '8' is 2, 3, and **4**.



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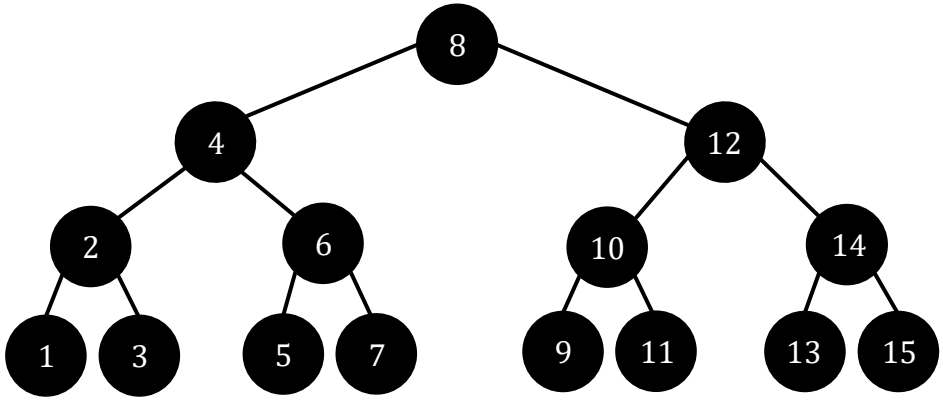


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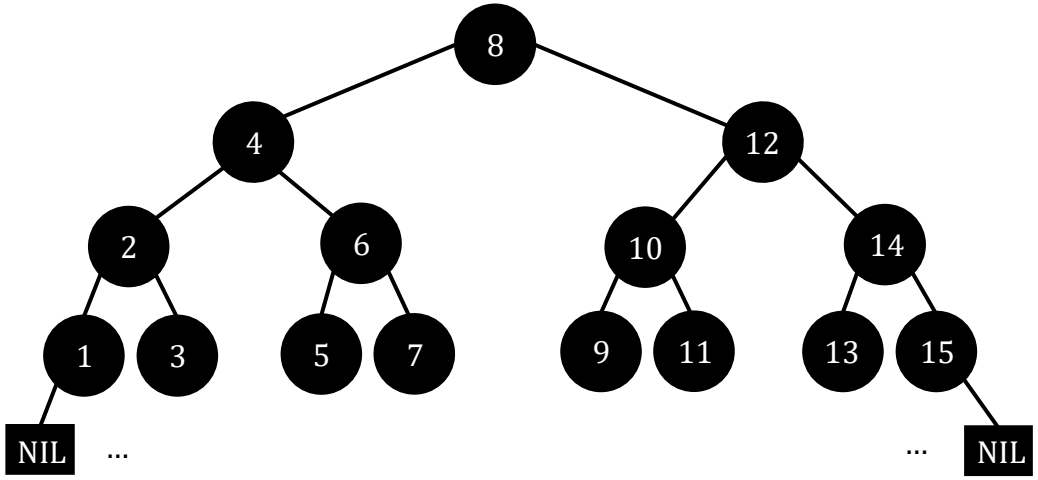


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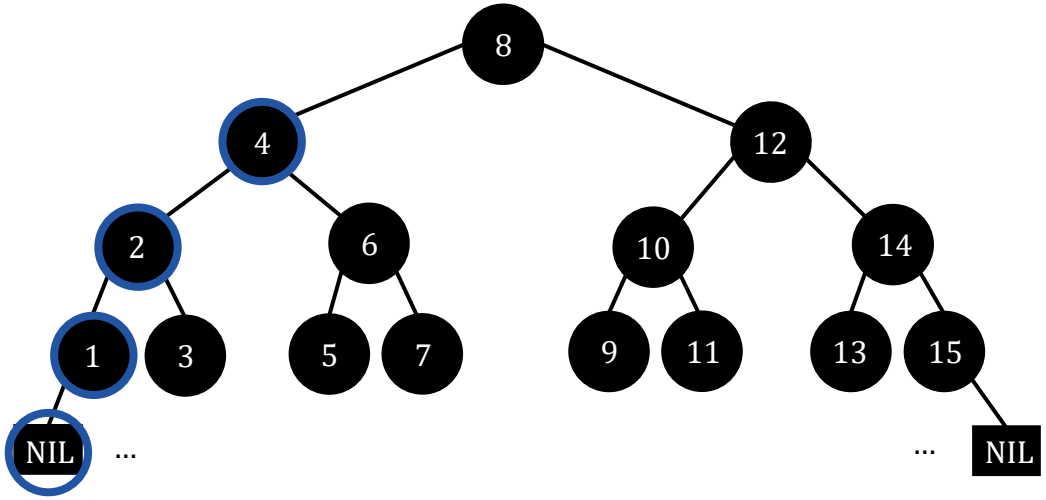


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Operations: Insertion & Deletion

Balancing a Red-Black Tree

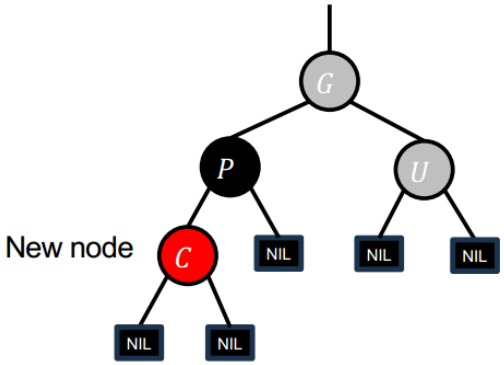
- When a Red-Black Tree is updated by insert or delete operations, it may violate the red-black tree properties.
- To restore the properties, one might **(1) change the colors of some nodes (recoloring)** or **(2) change the structure (rotation)**.

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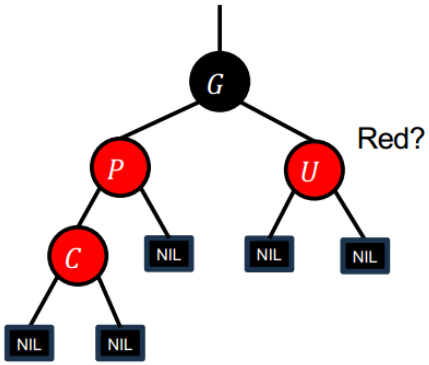
Operations: Insertion

Inserting a key *w* into *T*

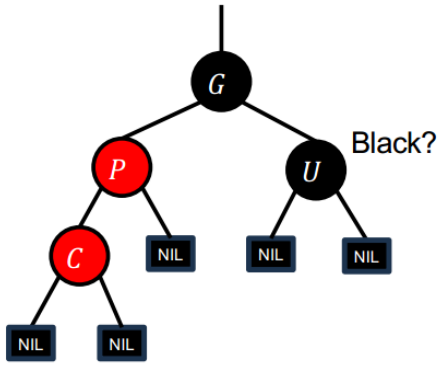
- [1] Perform standard BST insertion and make the color of new node as **Red**
- [2] If parent is **Black**, then do nothing!
- [3] If parent is **Red**, then do the following:
  - [3-a] If 'uncle' node U is **Red**, then **'Recolor'**
  - [3-b] If 'uncle' node U is **Black**, then **'Rotate+ Recolor'**



(2): the parent is black



(3)-(a): the uncle is red



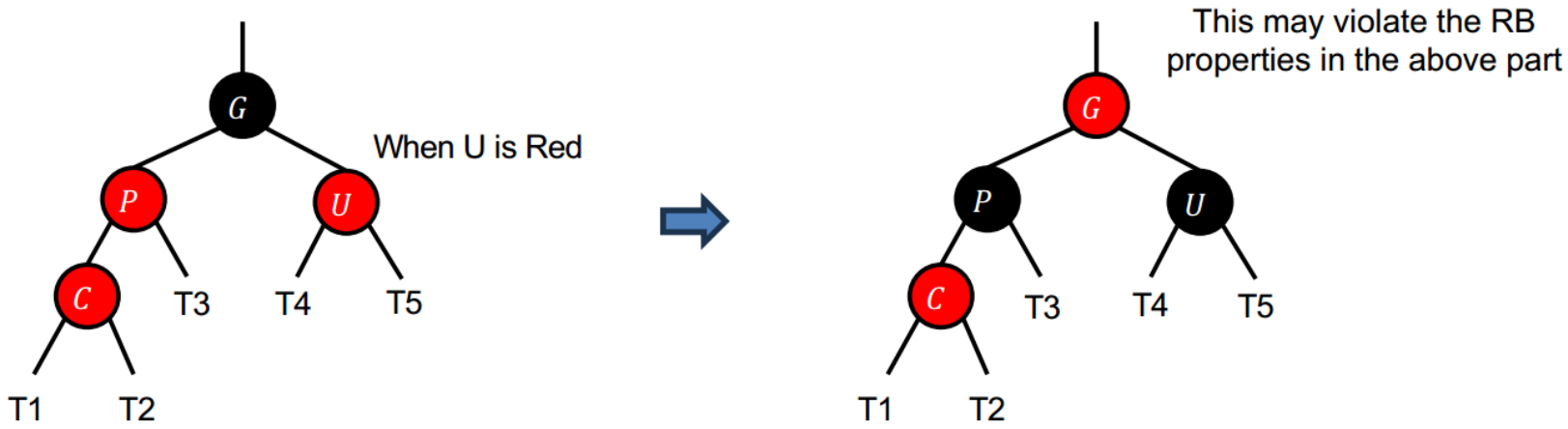
(3)-(b): the uncle is black

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Operations: Insertion

[CASE 1] U is **RED** : Recoloring

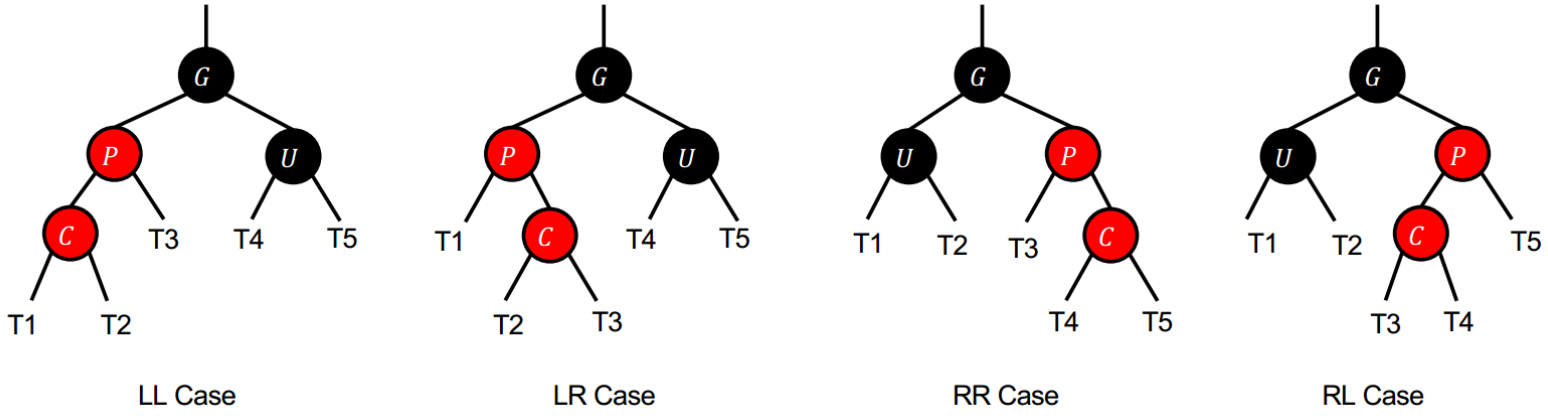
- [1] Change the colors of the parent P and the uncle U to **Black**
- [2] Change color of grandparent G as **Red**
- [3] Treat G as the New Node and repeat [2] and [3]



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Operations: Insertion

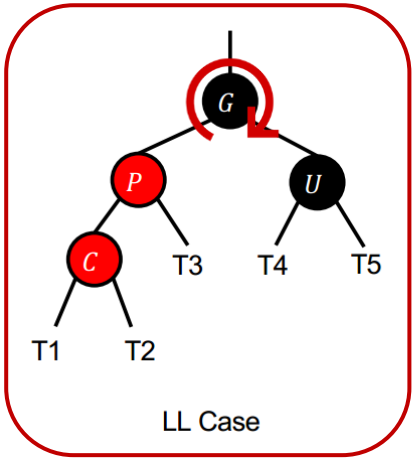
- [CASE 2] U is **Black** : Rotation + Recoloring
  - [1] Preserve the property of Binary Search Tree (left subtree <= root <= right subtree)
  - **[2] Rotate according to four scenarios : LL, LR, RR, RL**

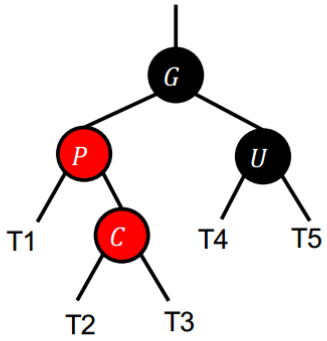


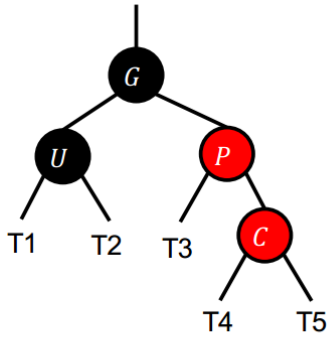
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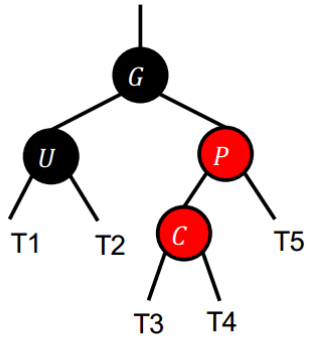
Operations: Insertion

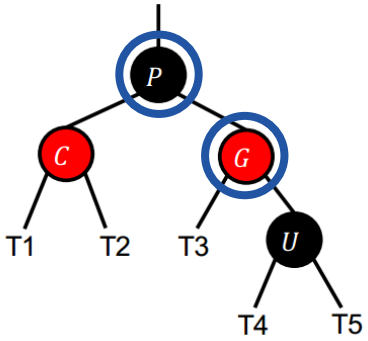
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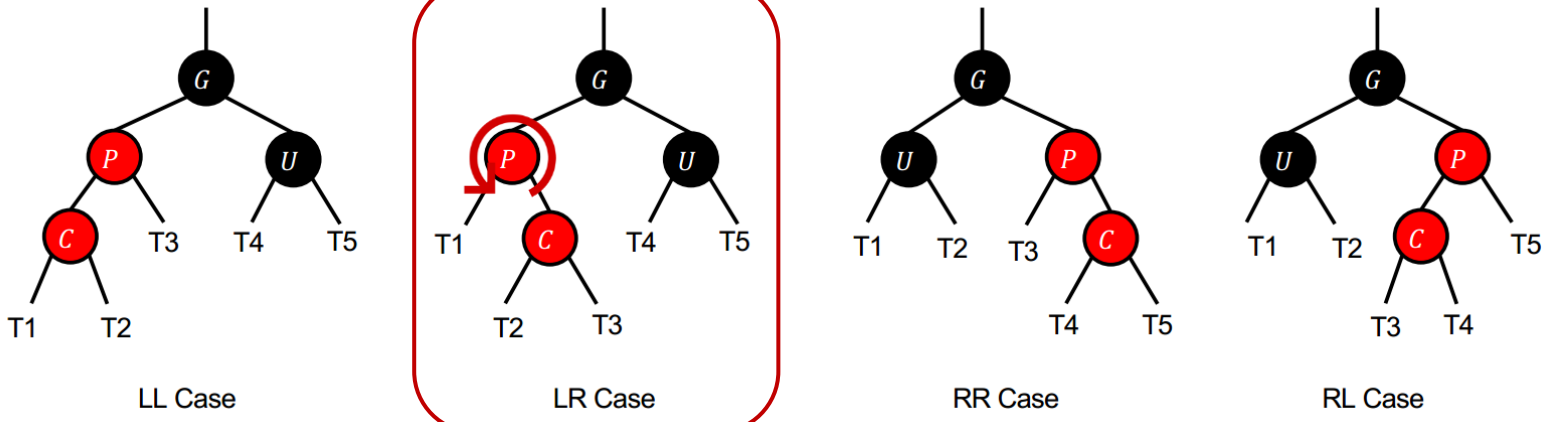


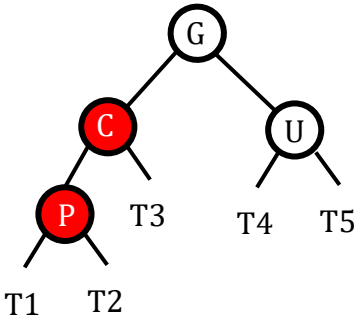


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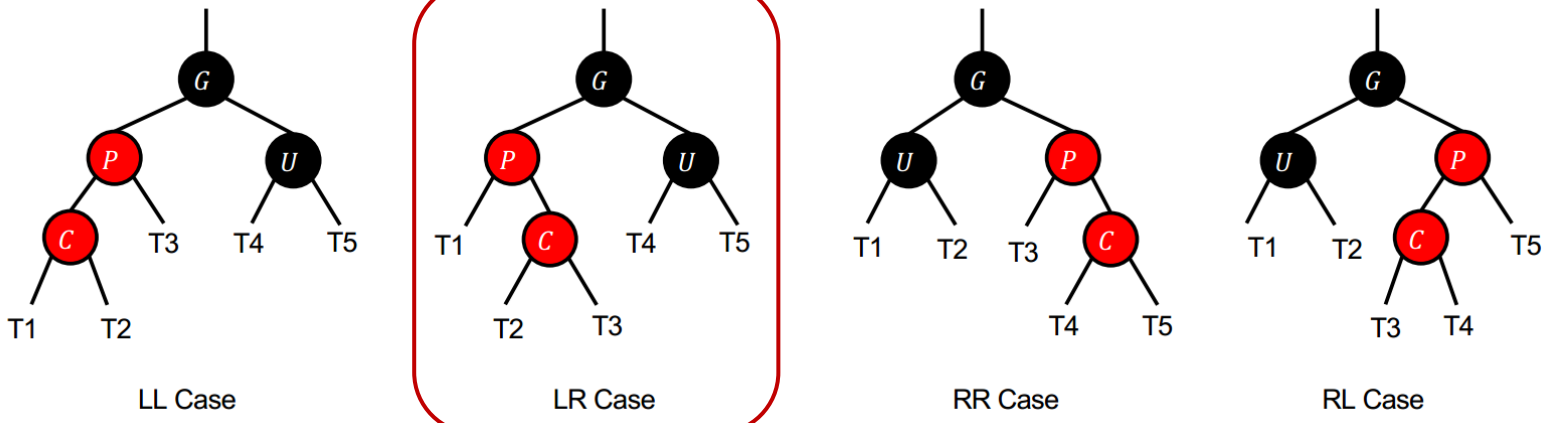


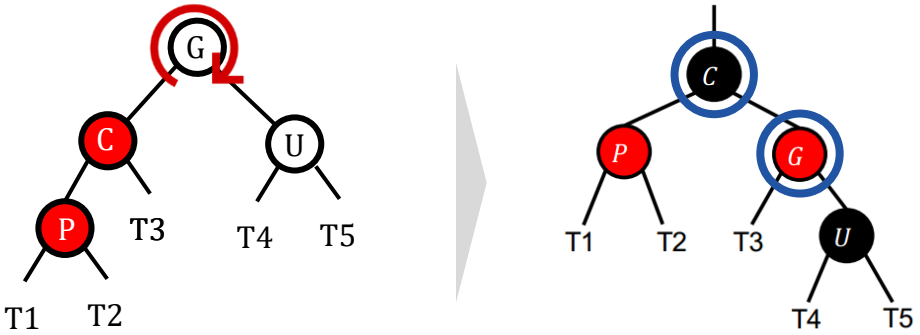


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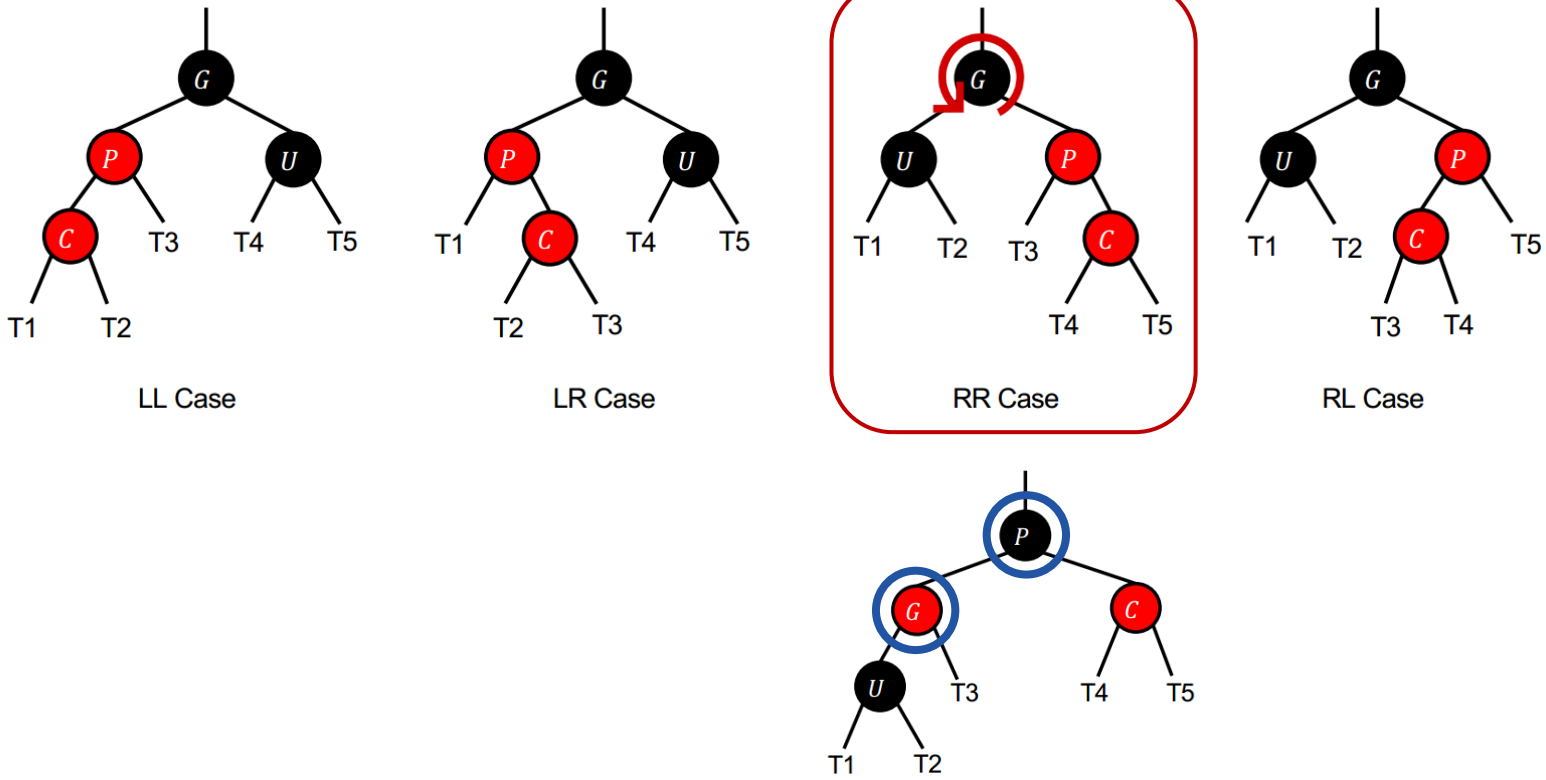




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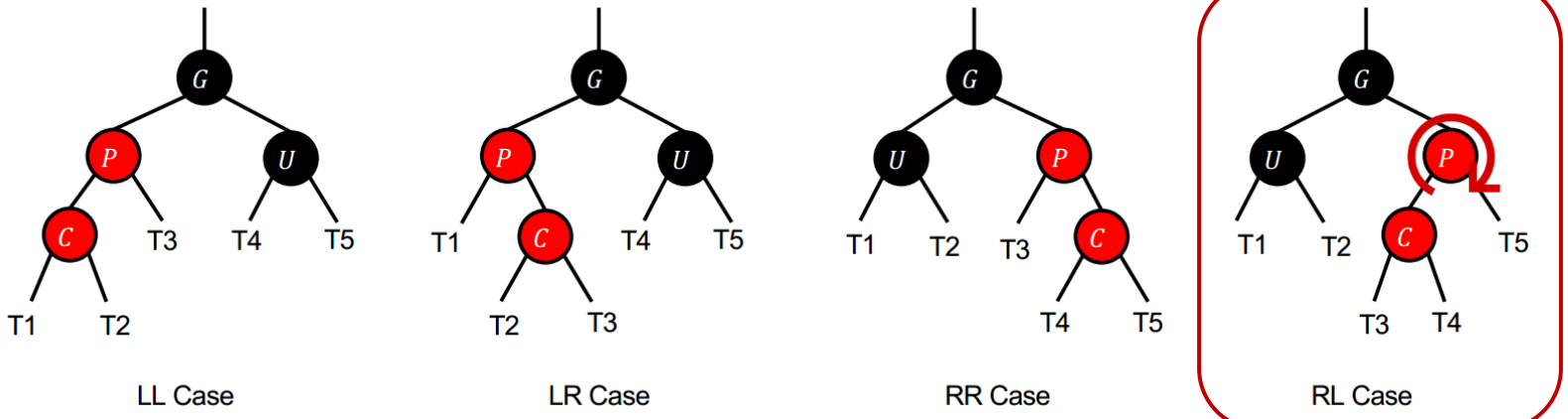
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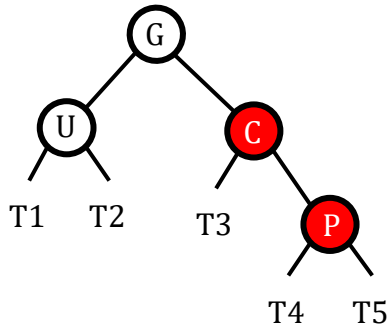


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  - [1] Preserve the property of Binary Search Tree (left subtree <= root <= right subtree)
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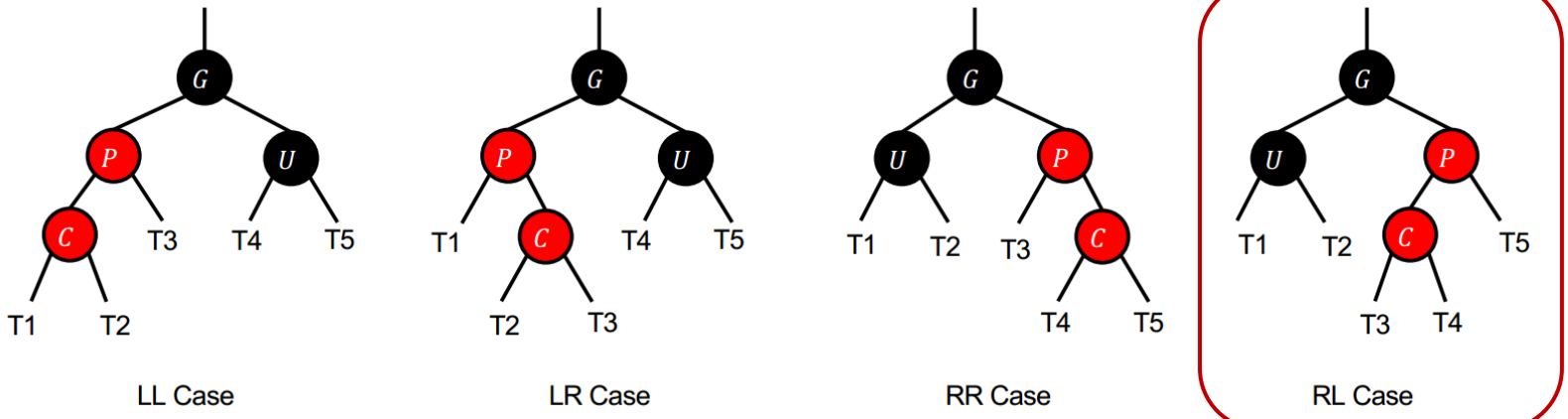


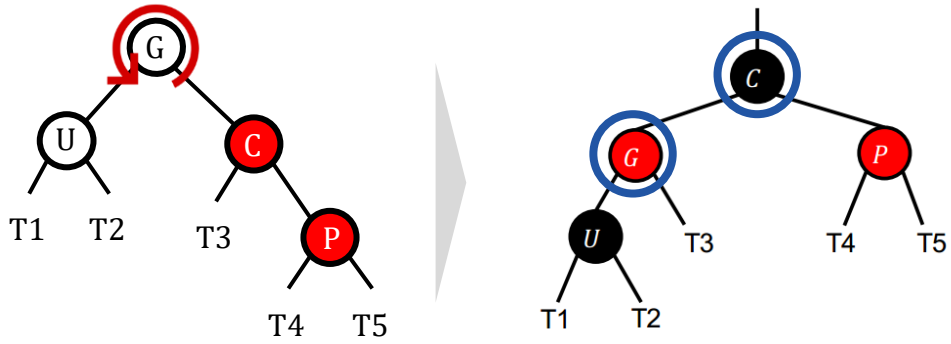


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Operations: Insertion

- [CASE 2] U is **Black** : Rotation + Recoloring
  - [1] Preserve the property of Binary Search Tree (left subtree <= root <= right subtree)
  - **[2] Rotate according to four scenarios : LL, LR, RR, RL**





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Why Red-Black Tree?

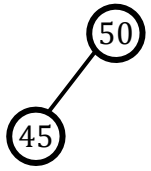
- Worst case in BST



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Why Red-Black Tree?

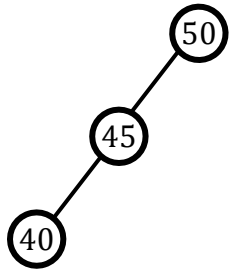
- Worst case in BST



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Why Red-Black Tree?

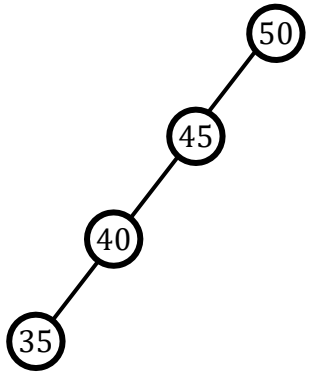
- Worst case in BST



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Why Red-Black Tree?

- Worst case in BST



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Why Red-Black Tree?

- Worst case in BST





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Why Red-Black Tree?

- Worst case in BST

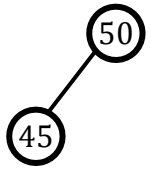


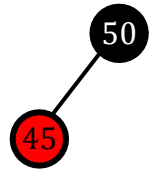


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Why Red-Black Tree?

- Worst case in BST

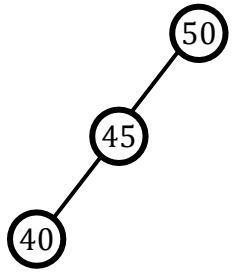


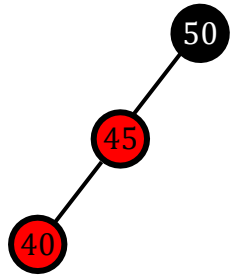


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Why Red-Black Tree?

- Worst case in BST

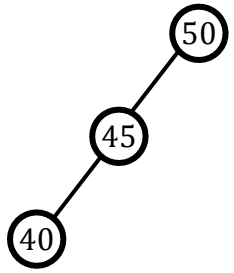


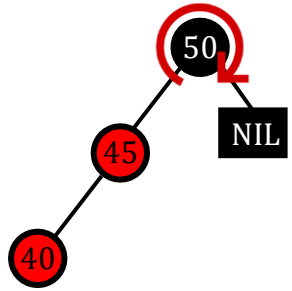


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Why Red-Black Tree?

- Worst case in BST

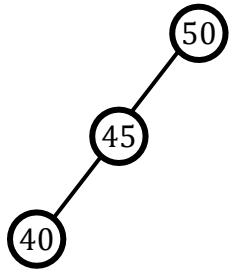


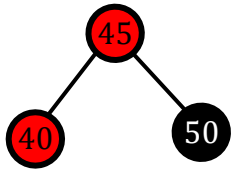


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Why Red-Black Tree?

- Worst case in BST

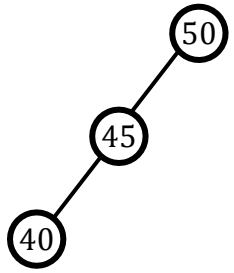


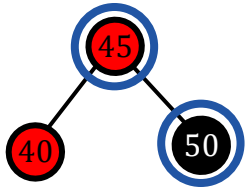


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Why Red-Black Tree?

- Worst case in BST

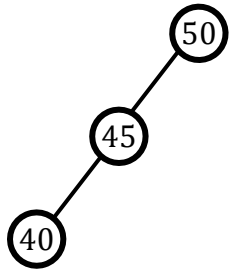


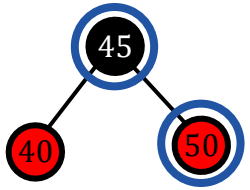


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Why Red-Black Tree?

- Worst case in BST

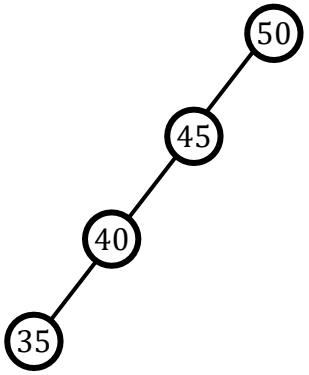


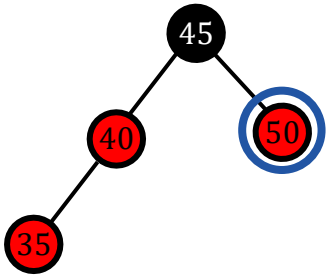


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Why Red-Black Tree?

- Worst case in BST

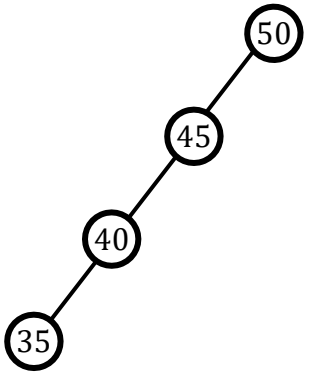


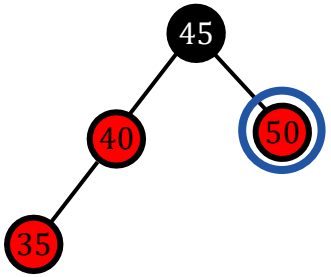


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Why Red-Black Tree?

- Worst case in BST

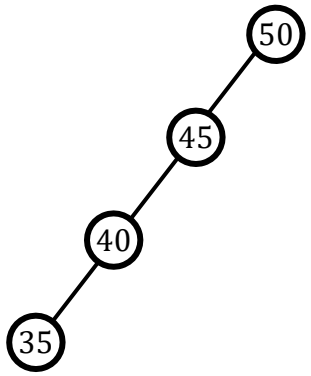


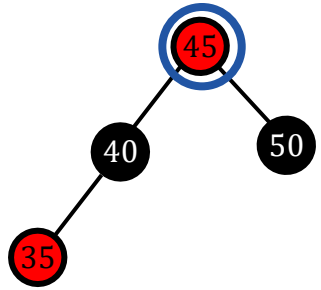


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Why Red-Black Tree?

- Worst case in BST



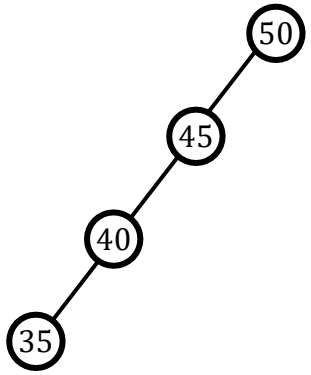


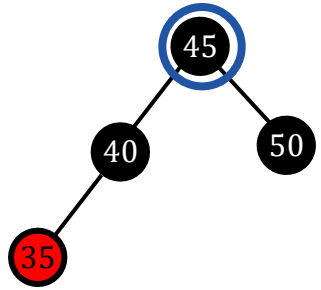
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Red-Black Tree

Why Red-Black Tree?

- Worst case in BST

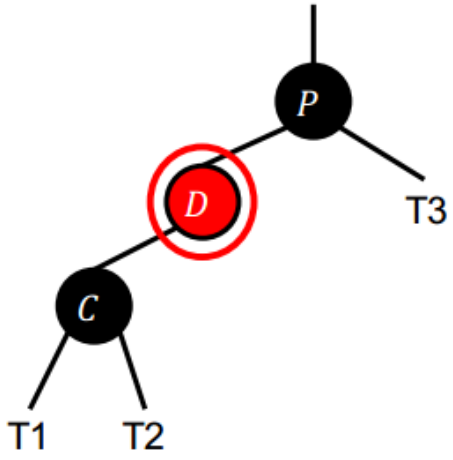




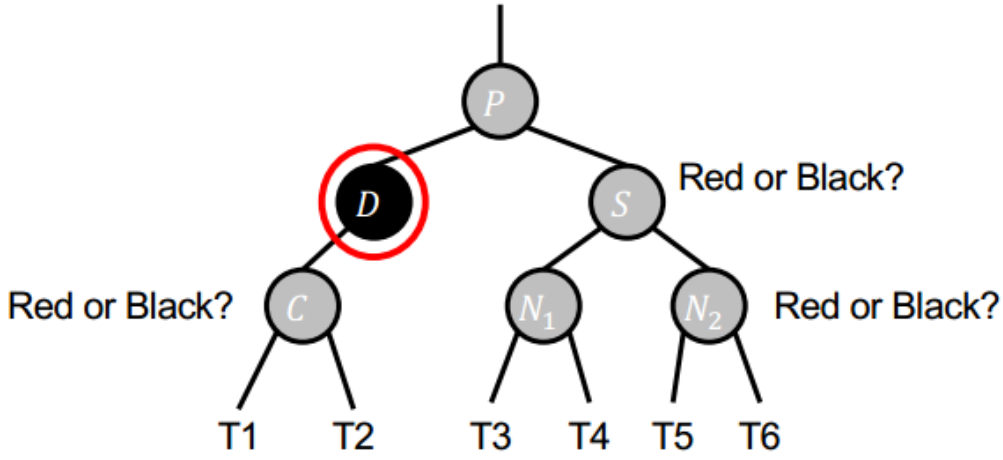
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Operations: Deletion

- Deleting D from BST T
  - [1] If the node has no child? Simply remove it!
  - [2] If the node has a single child? Replace the node with its child!
  - [3] If the node has two children? Replace the node with its successor!



(2): the node to be deleted is Red



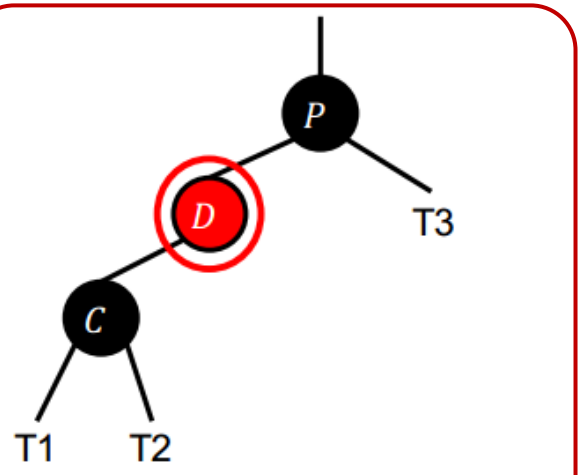
(3) The node to be deleted is Black

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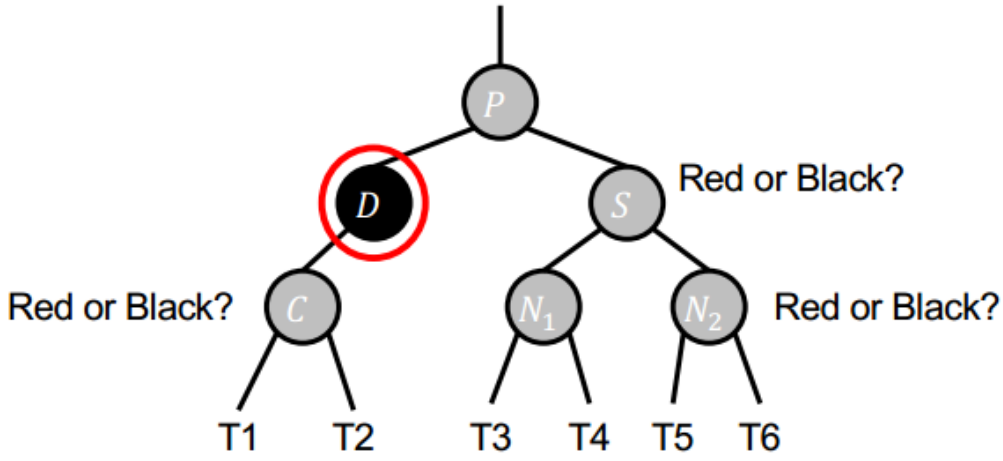
Operations: Deletion

- Deleting **Red** D from Red-Black Tree T
  - [1] If the node has no child? Simply remove it!
  - [2] If the node has a single child? Replace the node with its child!
  - [3] If the node has two children? Replace the node with its successor!

No violation if D is **Red**



(2): the node to be deleted is Red

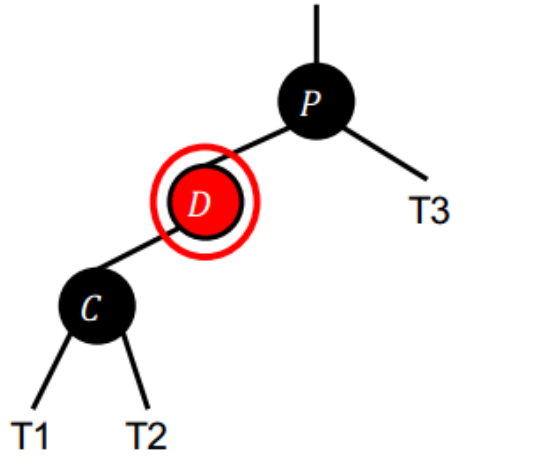


(3) The node to be deleted is Black

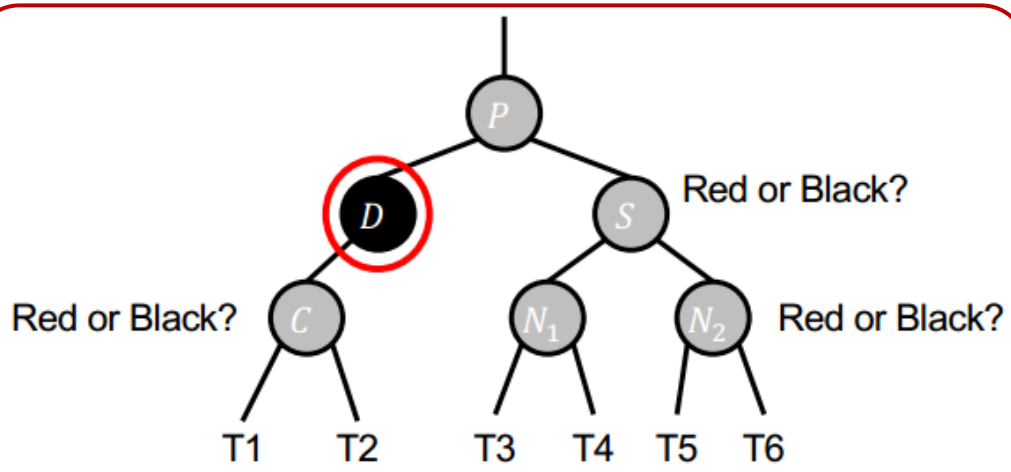
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Operations: Deletion

- Deleting **Black** D from Red-Black Tree T
  - [CASE 1] Successor C is Red
  - [CASE 2] Successor C is Black



(2): the node to be deleted is Red



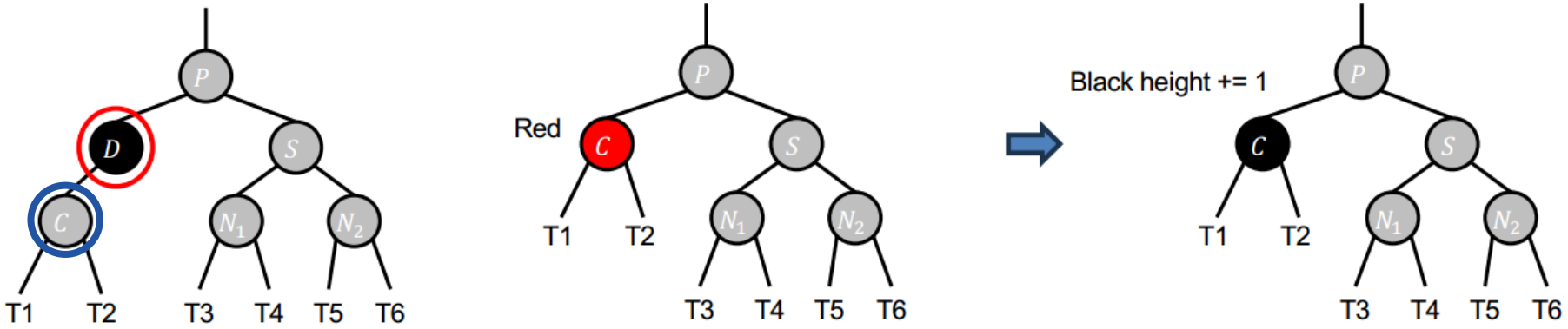
(3) The node to be deleted is Black

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Operations: Deletion

- Deleting **Black** *D* from Red-Black Tree *T*
  - **[CASE 1] Successor C is Red**
  - [CASE 2] Successor C is Black

- **Solution: Recoloring**
  - Change the color of node C as **Black**

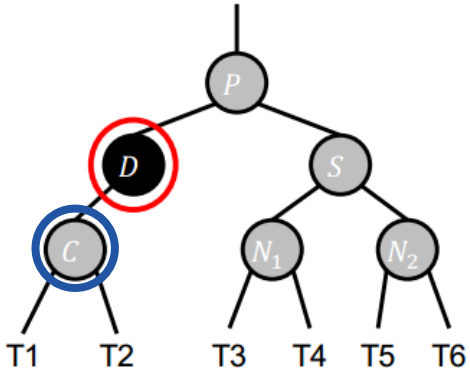


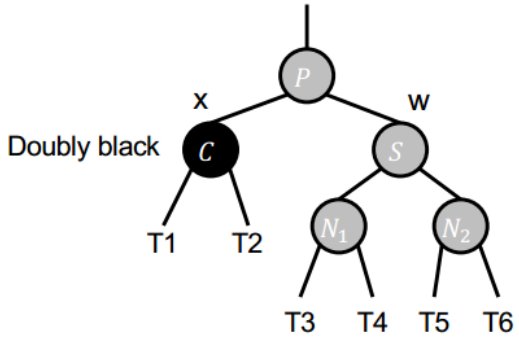
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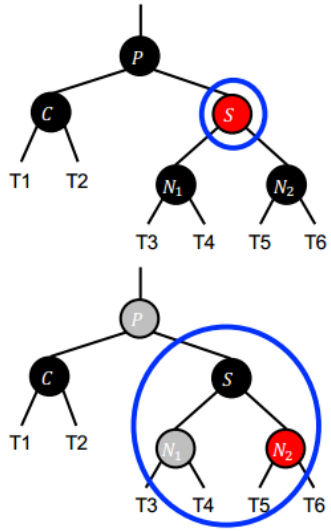
Operations: Deletion

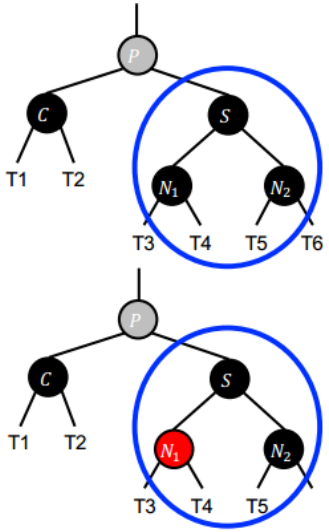
- Deleting **Black** *D* from Red-Black Tree *T*
  - [CASE 1] Successor C is Red
  - **[CASE 2] Successor C is Black**

- **Solution: Rotation + Recoloring**
  - Four cases







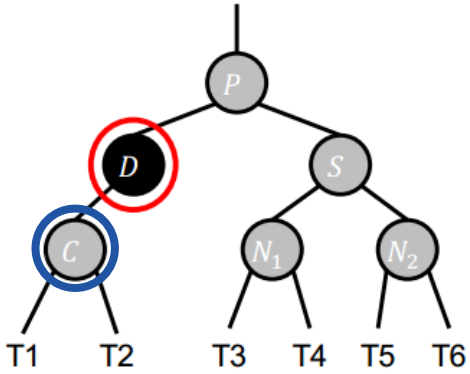


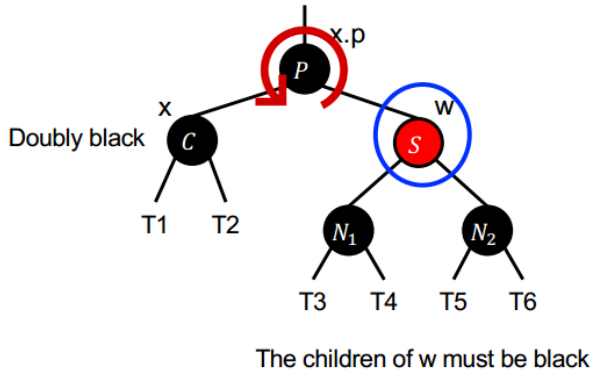
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Operations: Deletion

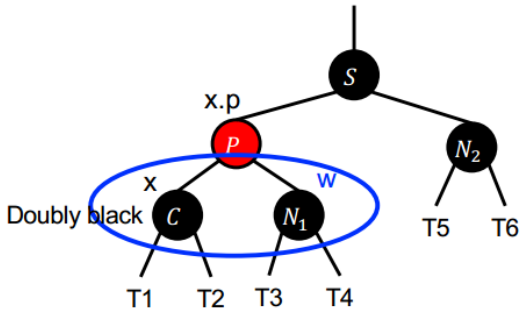
- Deleting **Black** *D* from Red-Black Tree *T*
  - [CASE 1] Successor C is Red
  - **[CASE 2] Successor C is Black**

- **[CASE 2-1] Sibling is Red**
  - Solution: Left Rotation + Recoloring





The children of w must be black



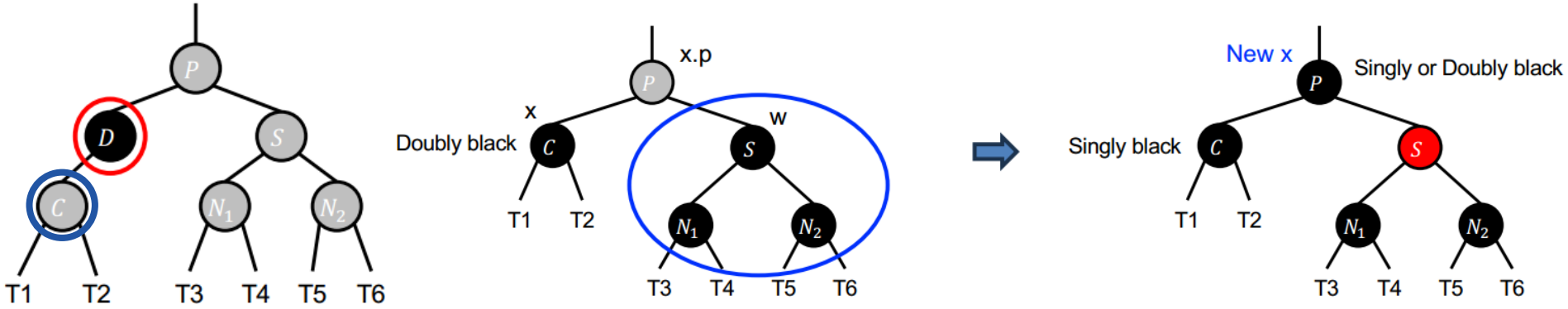
Convert Case 2-1 to into Case 2-2, 2-3, or 2-4.

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Operations: Deletion

- Deleting **Black** *D* from Red-Black Tree *T*
  - [CASE 1] Successor C is Red
  - **[CASE 2] Successor C is Black**

- **[CASE 2-2] Sibling is Black + children are Black**
  - Solution: Recoloring

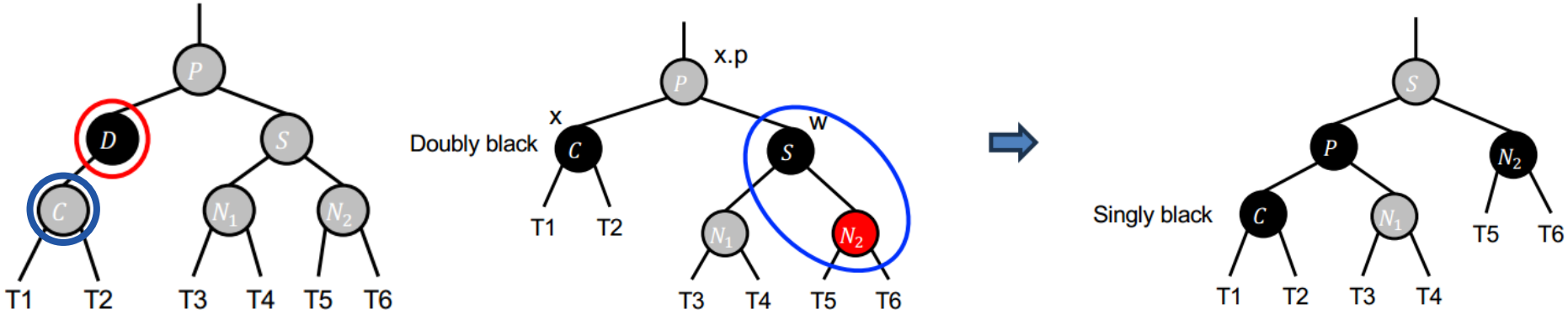


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Operations: Deletion

- Deleting **Black** *D* from Red-Black Tree *T*
  - [CASE 1] Successor C is Red
  - **[CASE 2] Successor C is Black**

- **[CASE 2-3] Sibling is Black + right child is Red**
  - Solution: Left Rotation + Recoloring (P and N2 to **Black**)

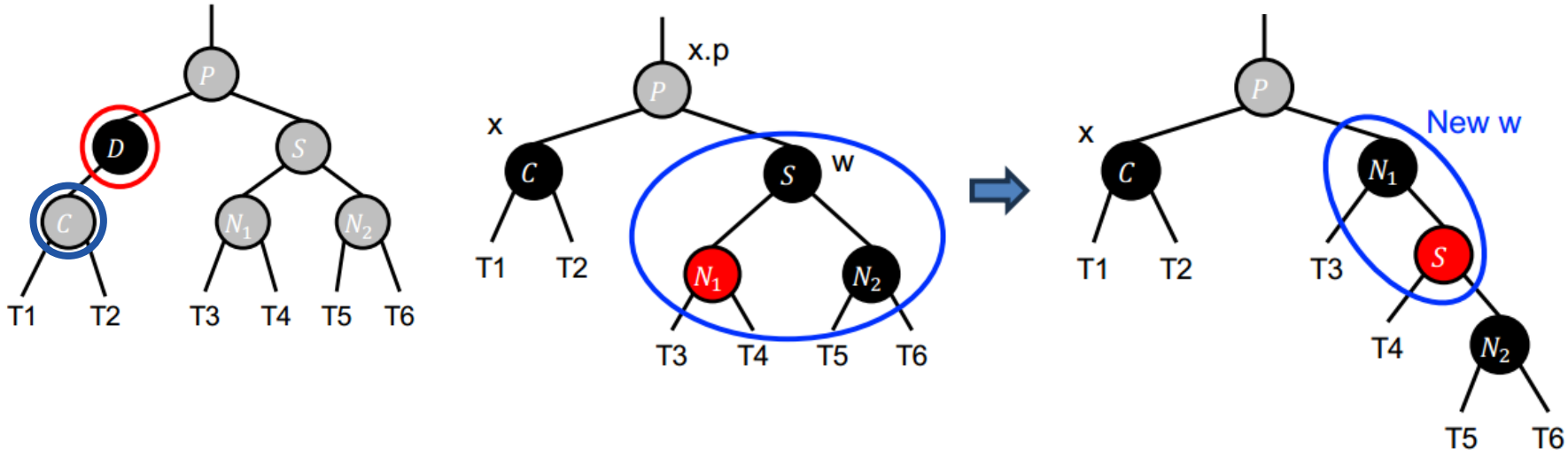


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Operations: Deletion

- Deleting **Black** *D* from Red-Black Tree *T*
  - [CASE 1] Successor C is Red
  - **[CASE 2] Successor C is Black**

- **[CASE 2-4] Sibling is Black + left child is Red + right child is Black**
  - Solution: Right Rotation + Recoloring (N1 to **Black, S to Red**) + Left Rotation + Recoloring (P to **Black**, S to **Black**)

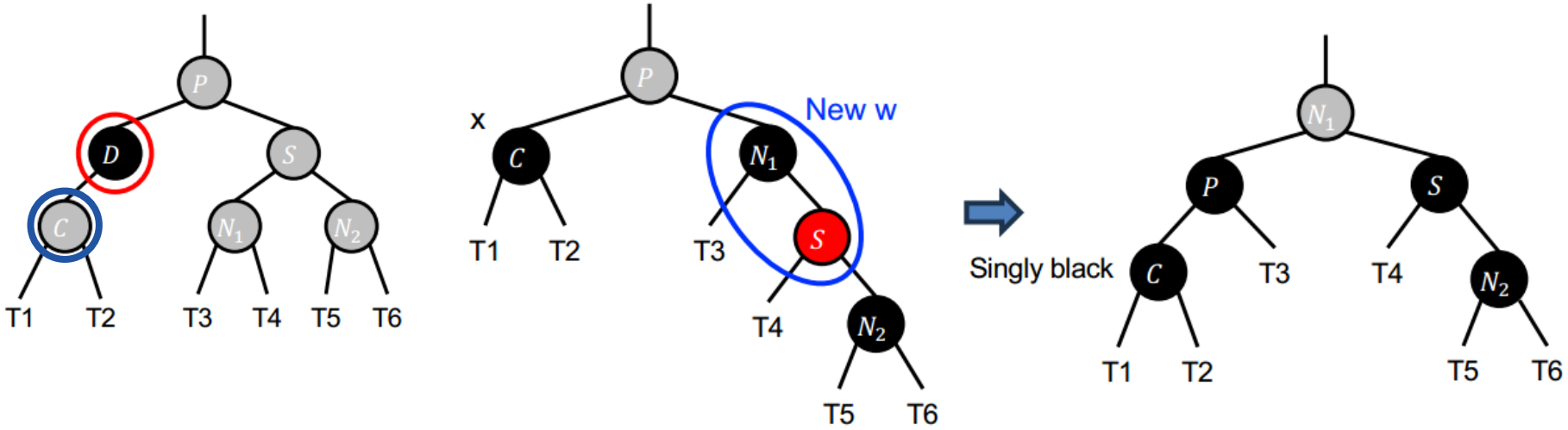


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Operations: Deletion

- Deleting **Black** *D* from Red-Black Tree *T*
  - [CASE 1] Successor C is Red
  - **[CASE 2] Successor C is Black**

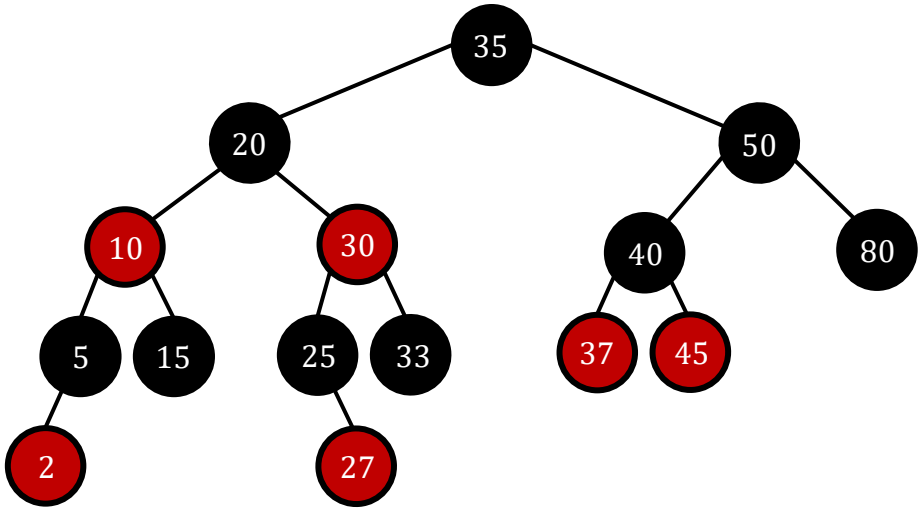
- **[CASE 2-4] Sibling is Black + left child is Red + right child is Black**
  - Solution: Right Rotation + Recoloring (N1 to **Black, S to Red**) + Left Rotation + Recoloring (P to **Black**, S to **Black**)



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Examples

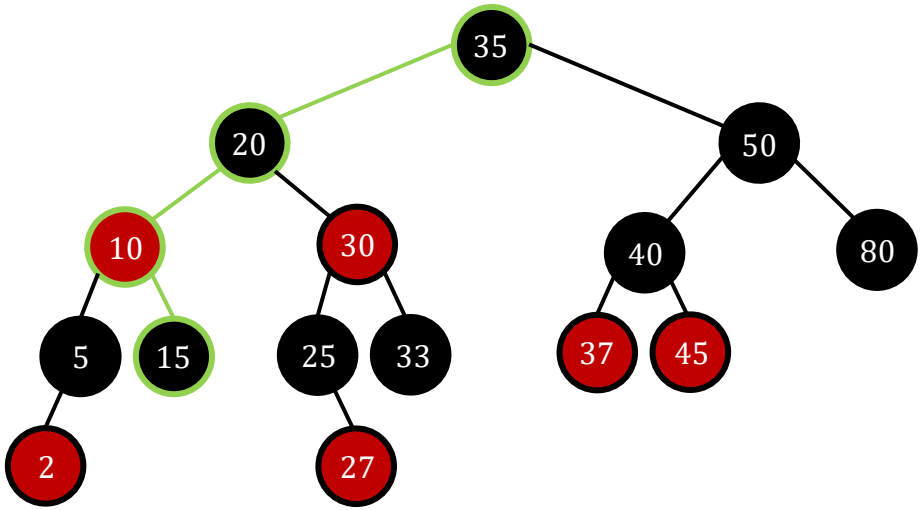
- Delete(15)



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Examples

- Delete(15)

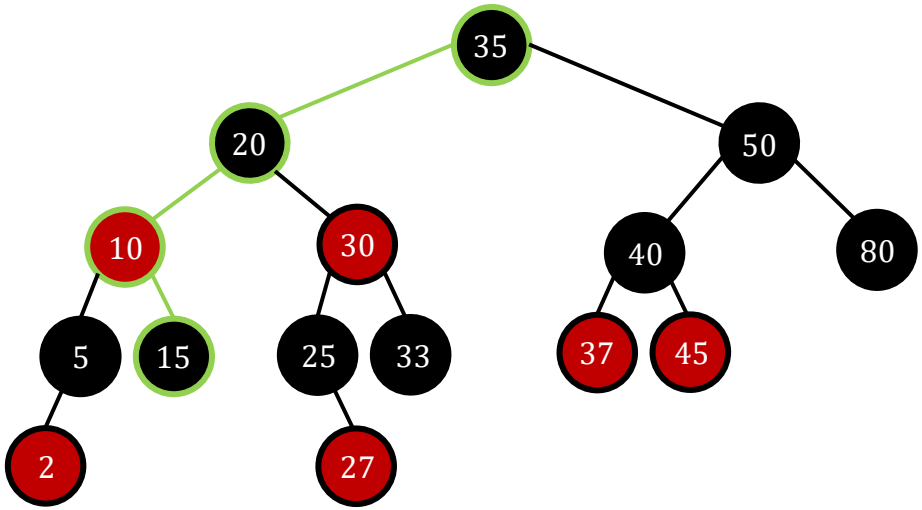


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Red-Black Tree

Examples

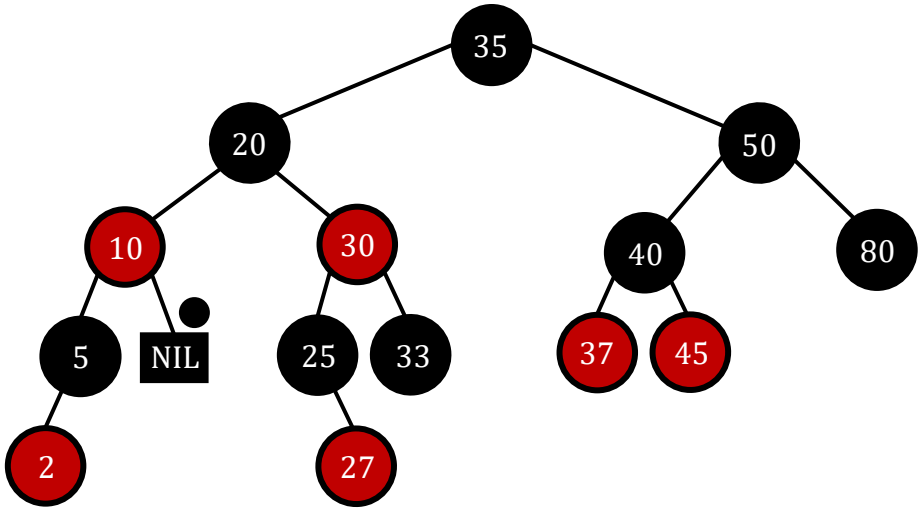
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Examples

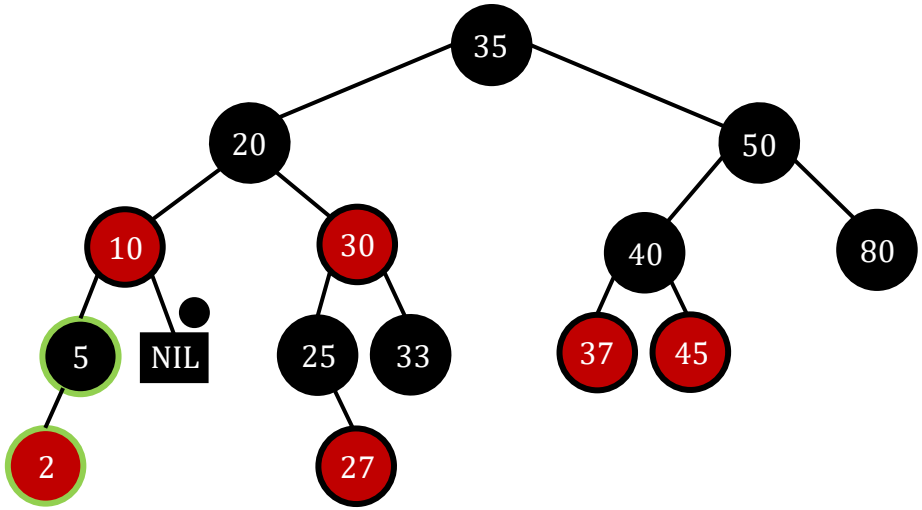
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Examples

- Delete(15)

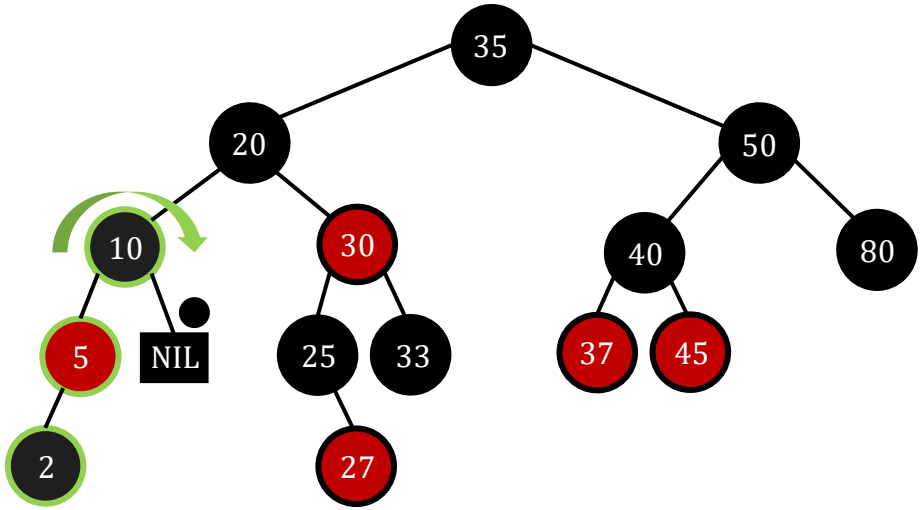


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Red-Black Tree

Examples

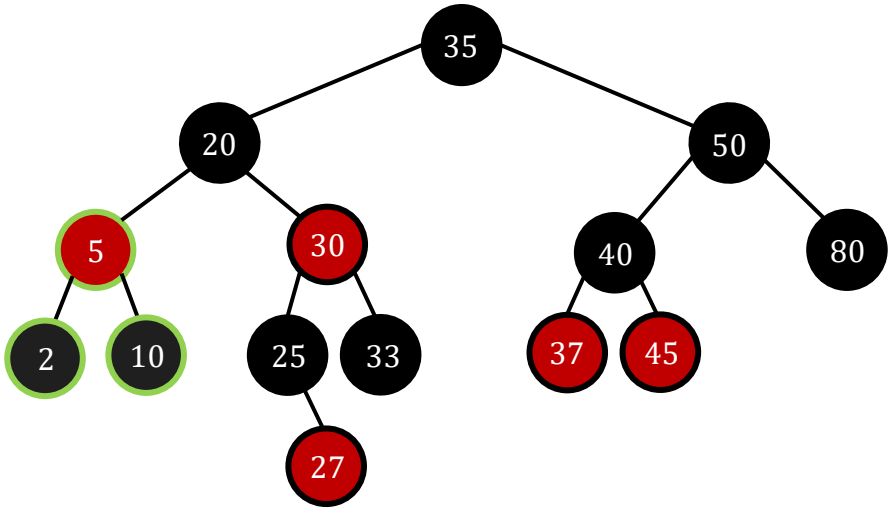
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Examples

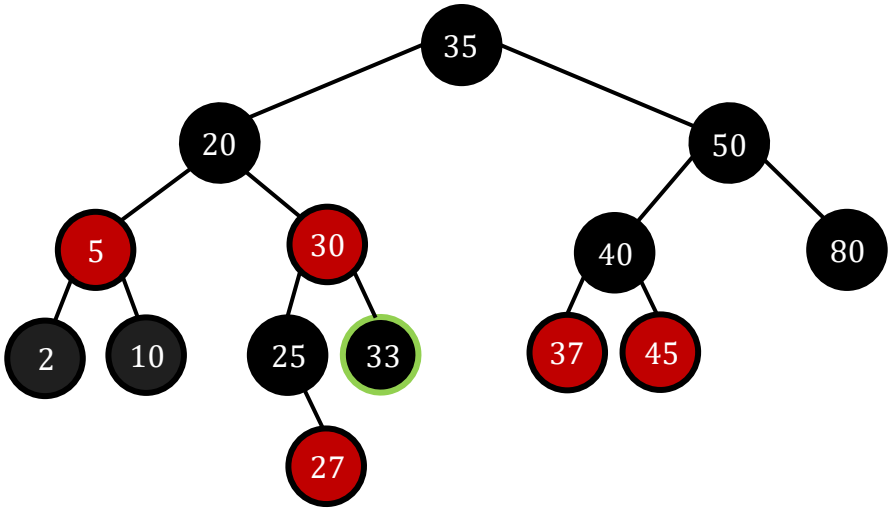
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Examples

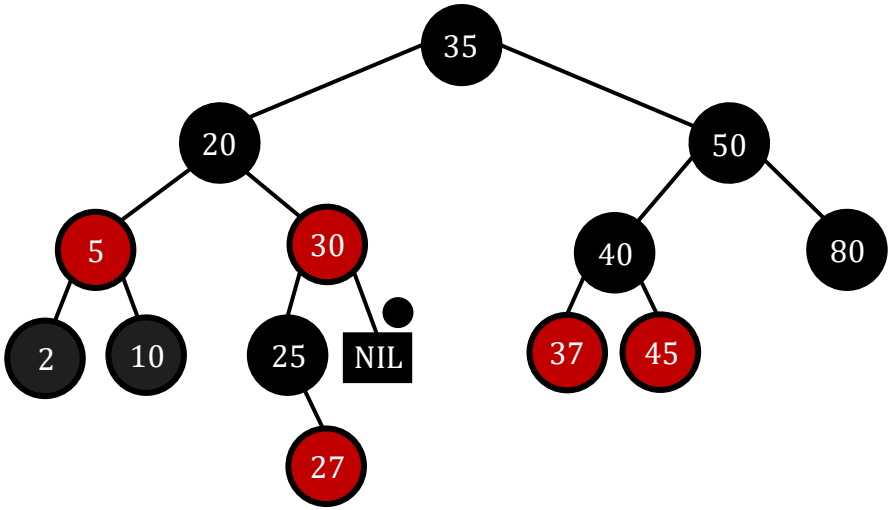
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Examples

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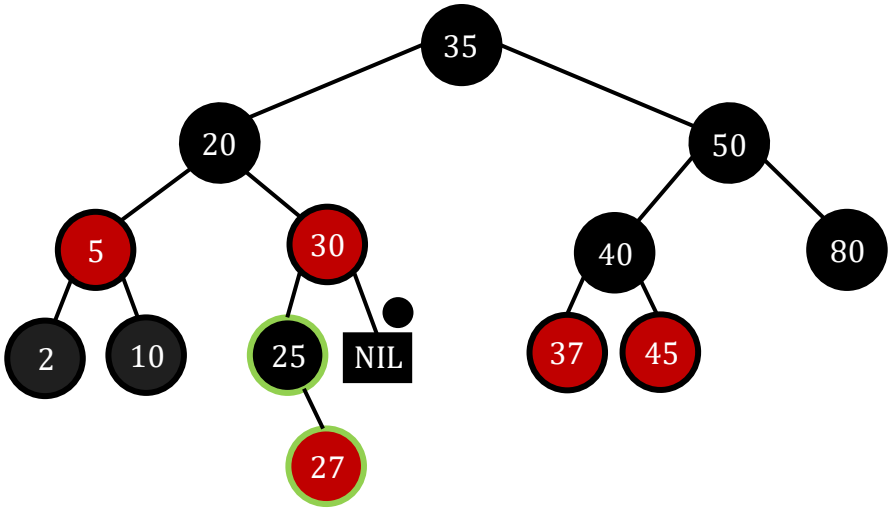


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Red-Black Tree

Examples

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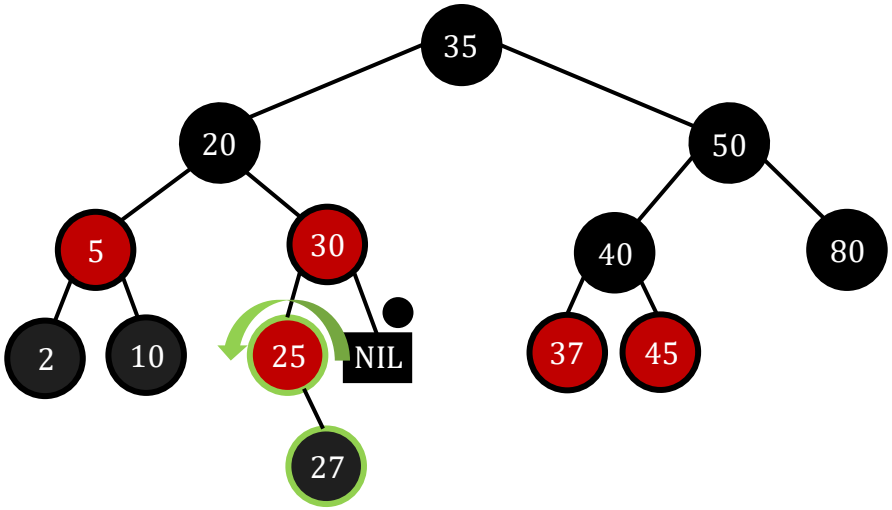


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Red-Black Tree

Examples

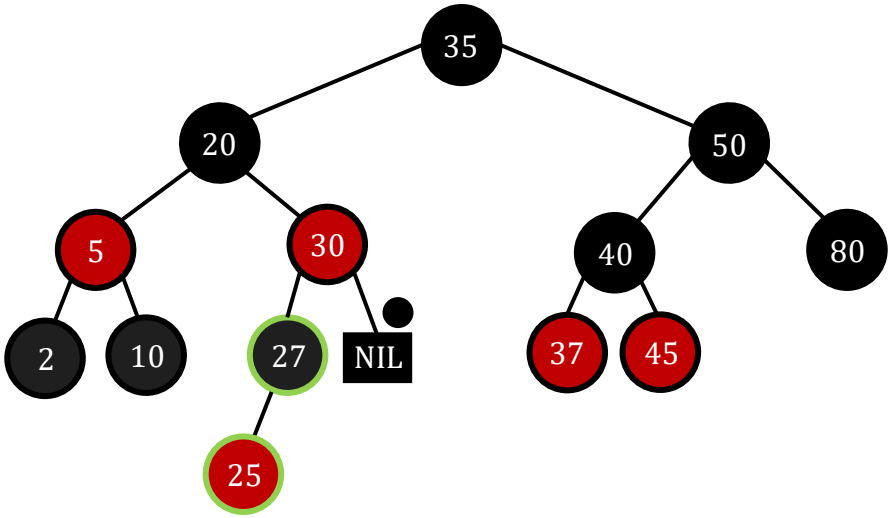
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Examples

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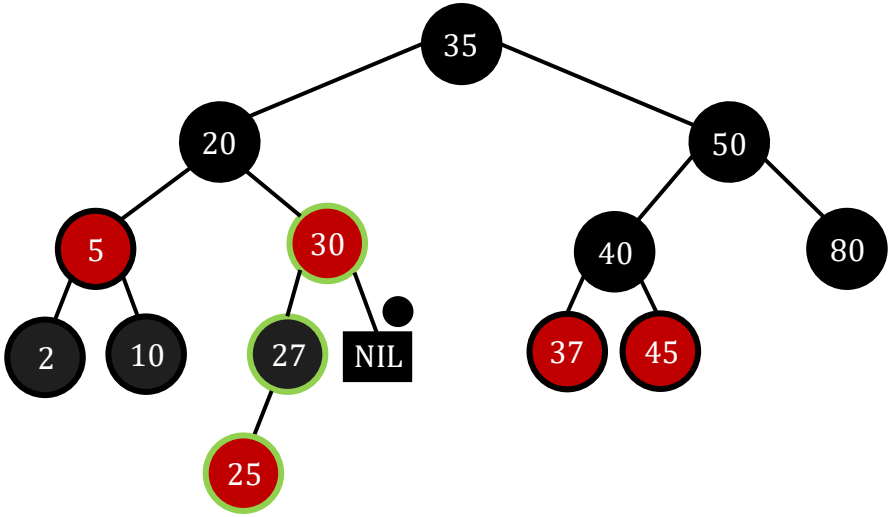


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Red-Black Tree

Examples

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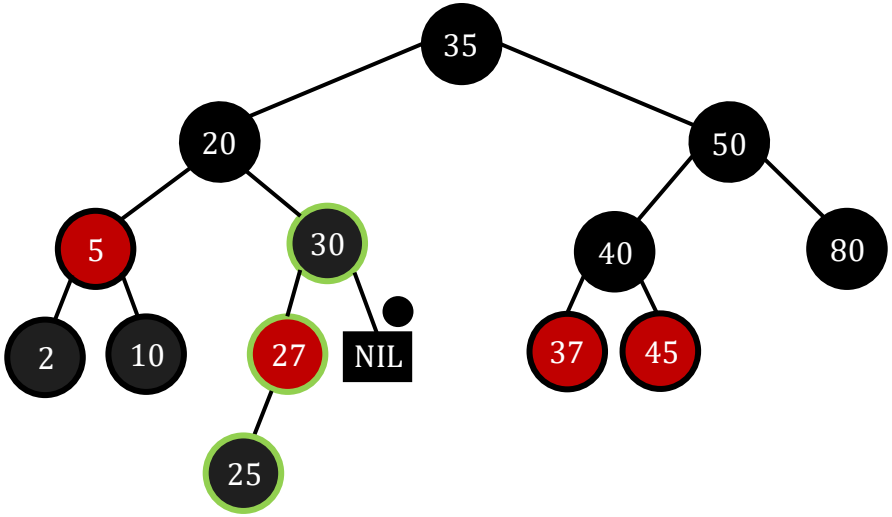


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Red-Black Tree

Examples

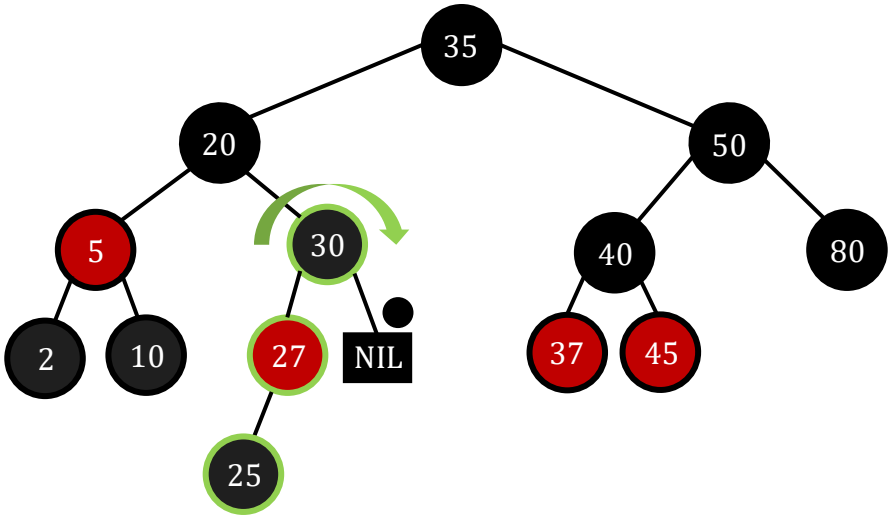
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Examples

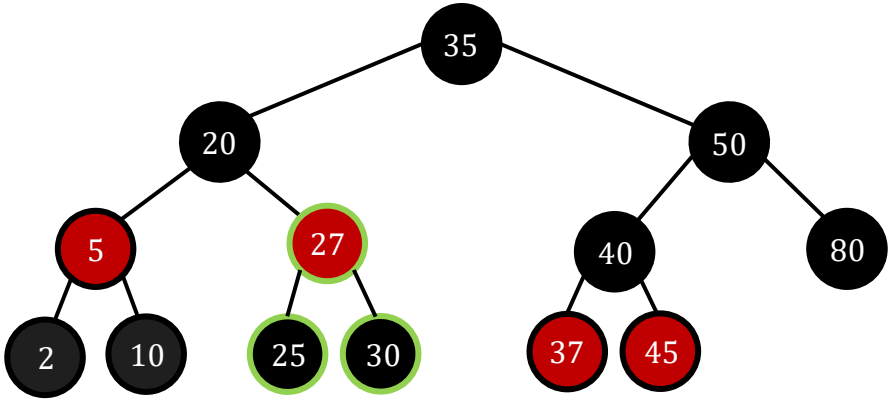
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Examples

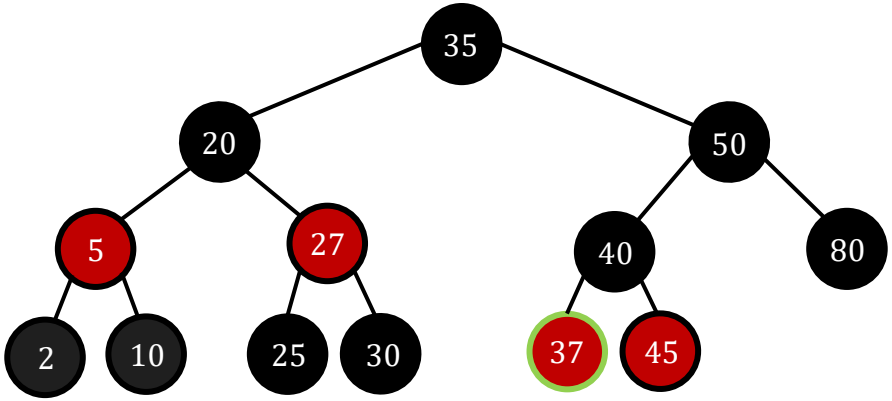
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Examples

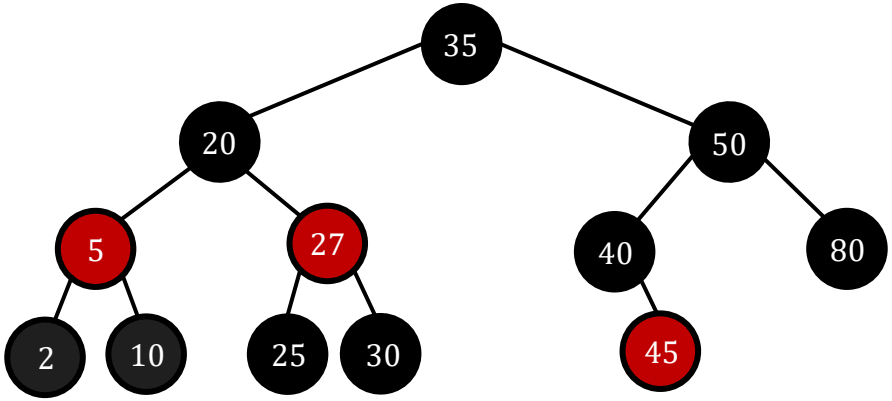
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Examples

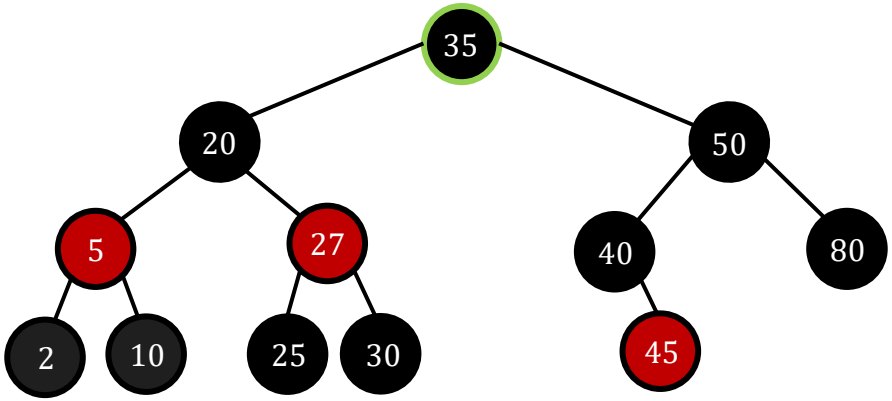
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Examples

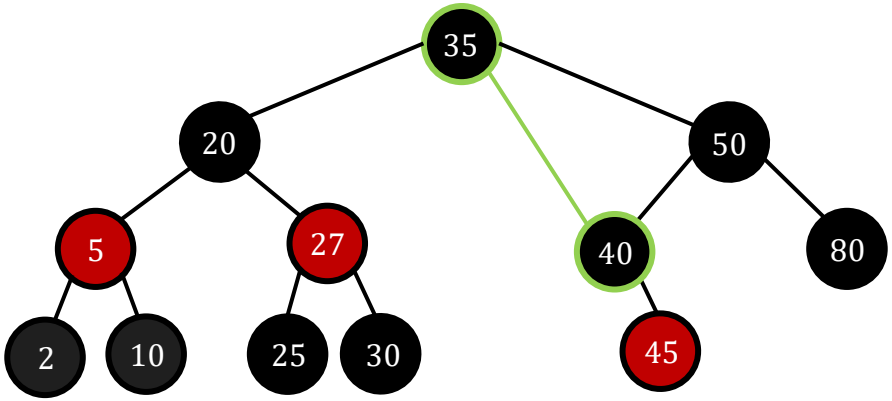
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Examples

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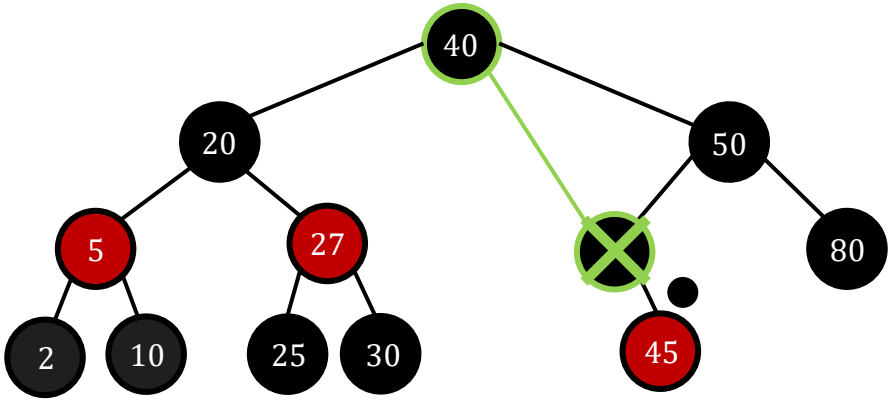


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Red-Black Tree

Examples

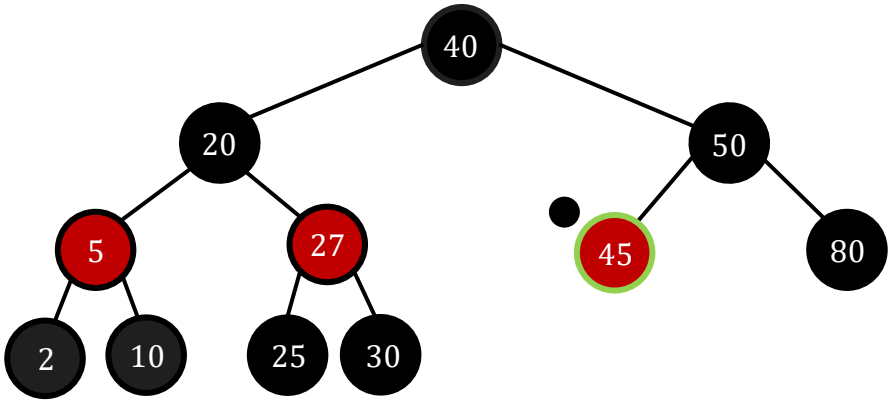
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Examples

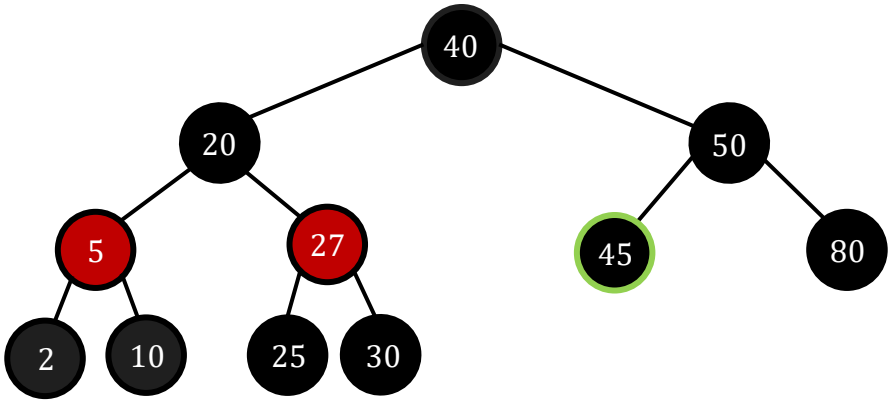
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Examples

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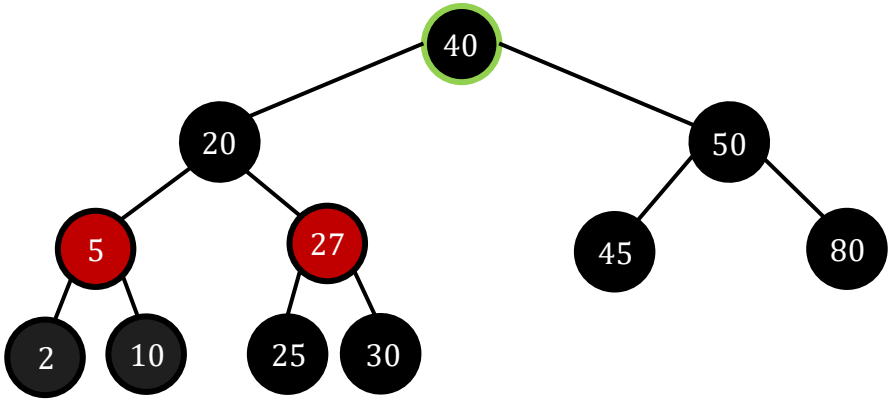


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Red-Black Tree

Examples

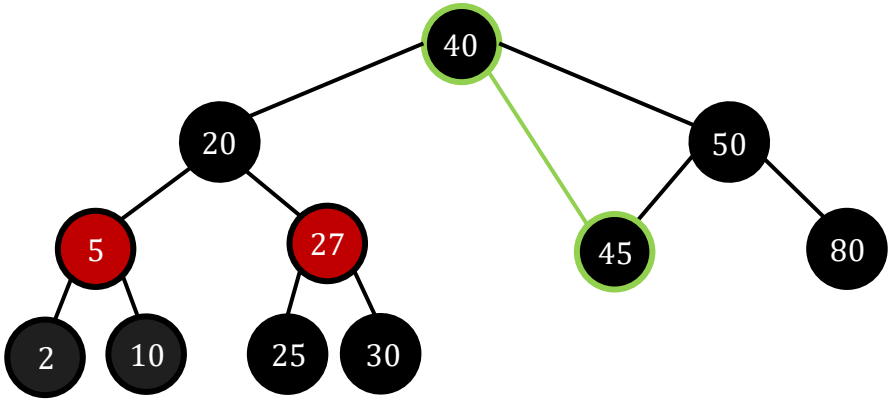
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Examples

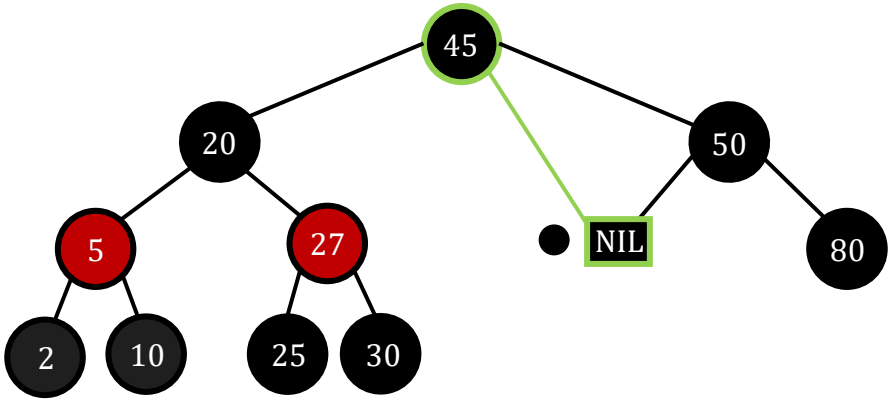
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Examples

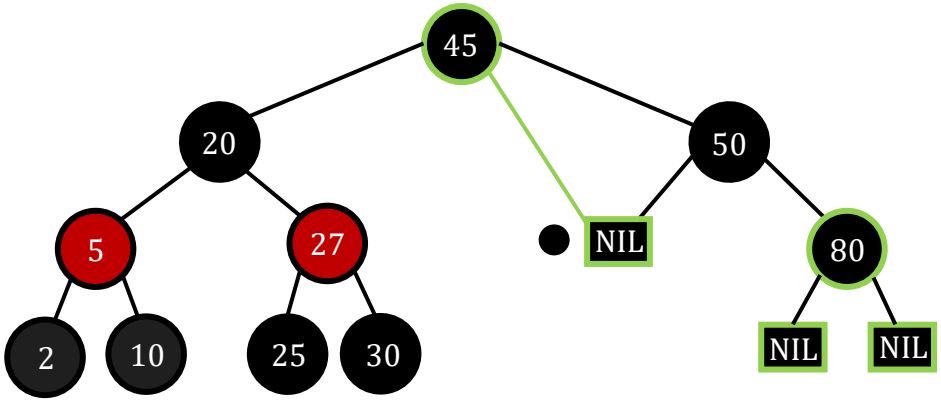
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Examples

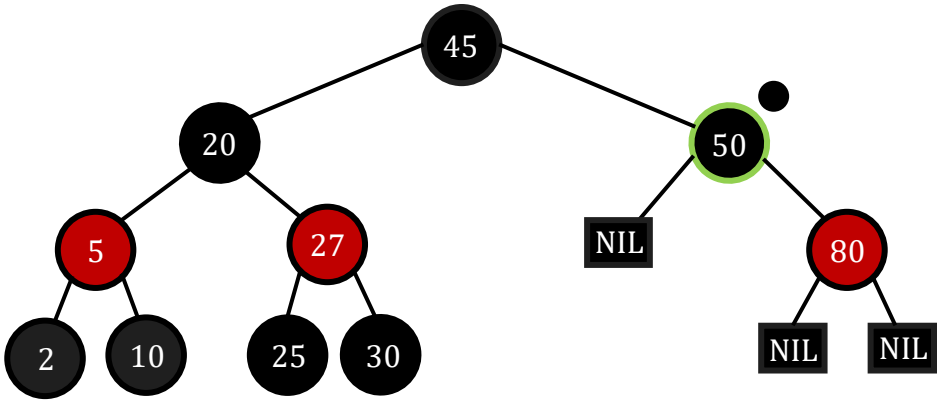
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Examples

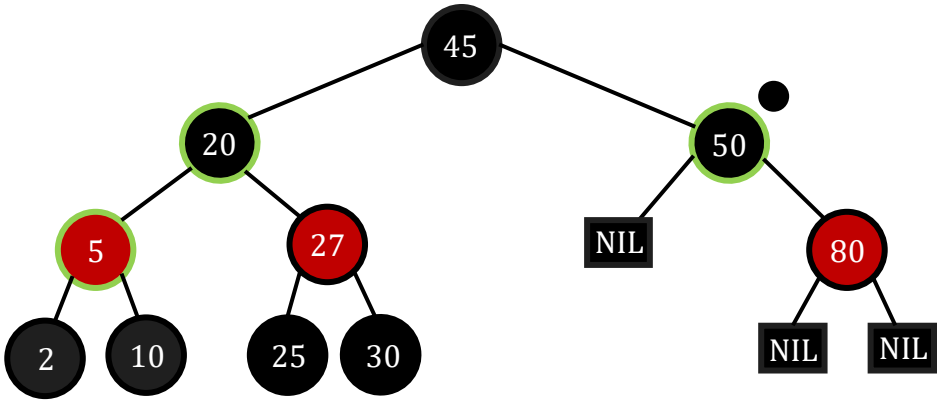
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Examples

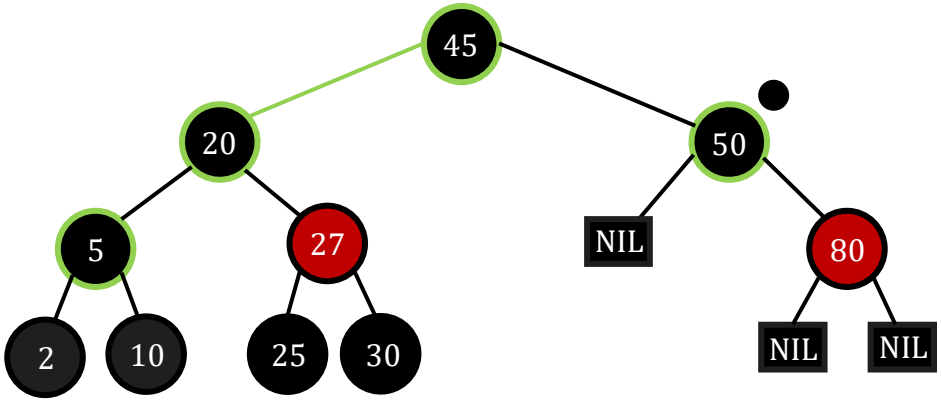
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Examples

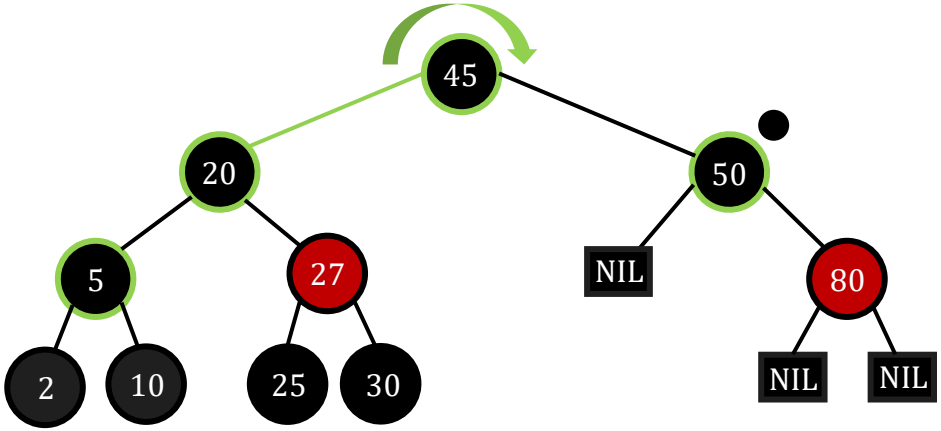
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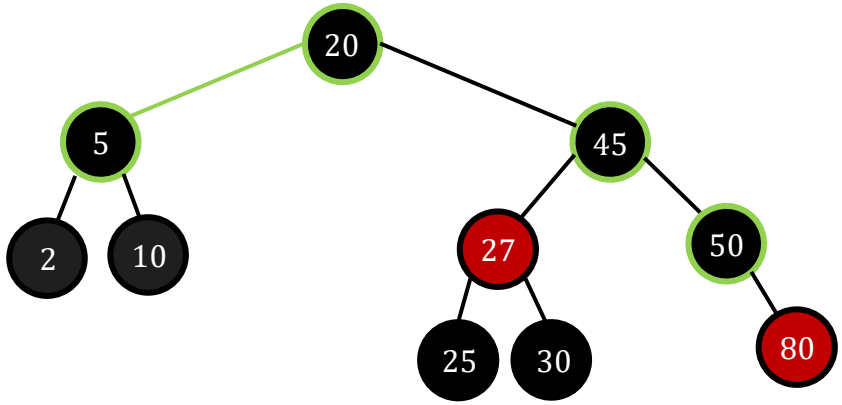
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Examples

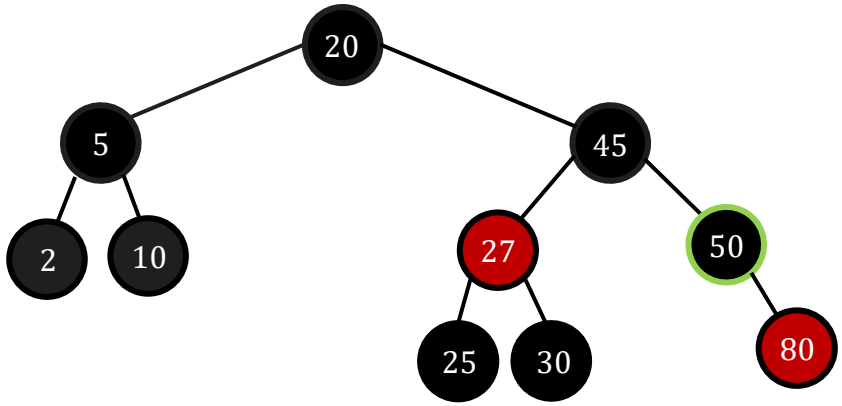
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Examples

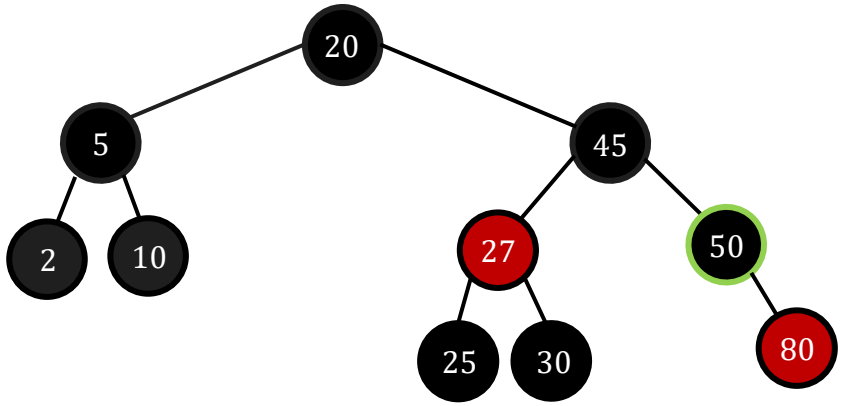
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Examples

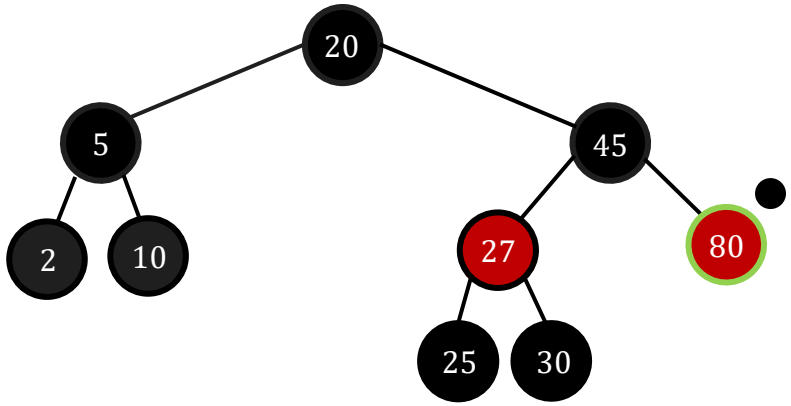
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Examples

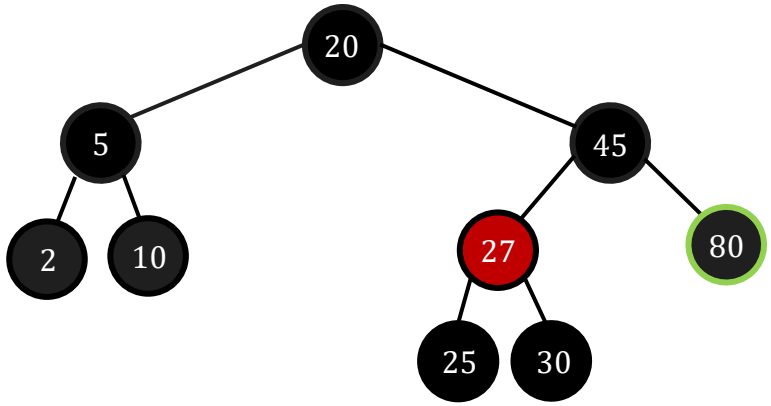
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Examples

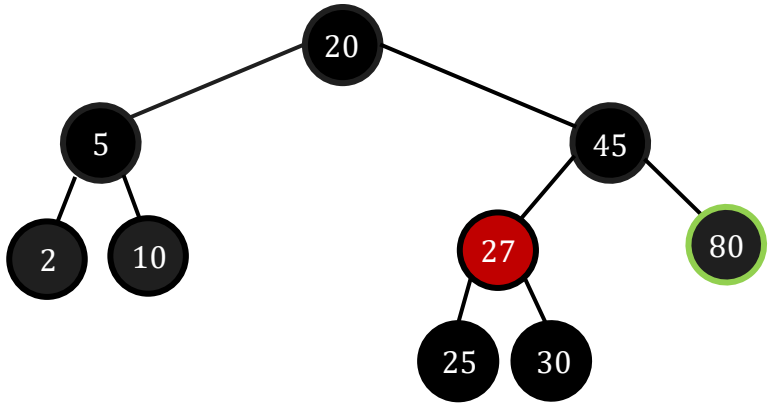
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Examples

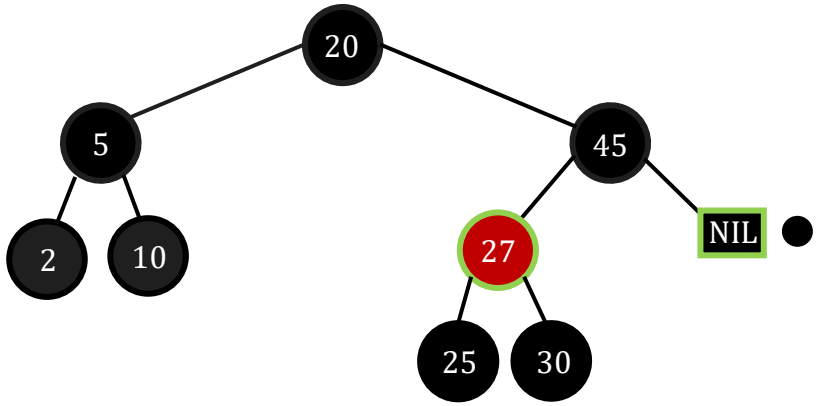
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Examples

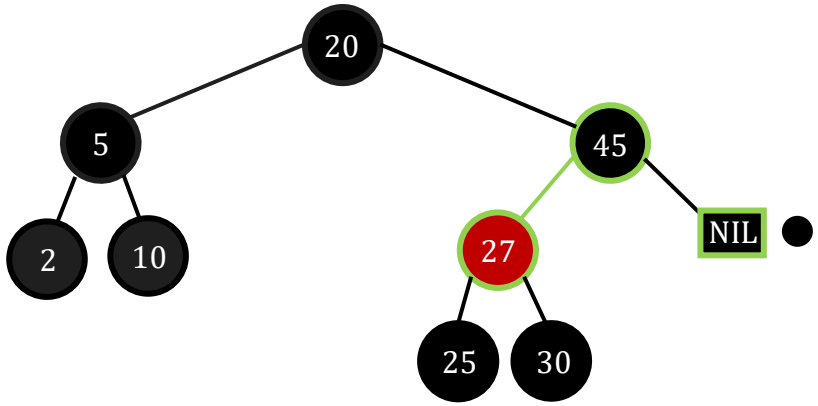
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Examples

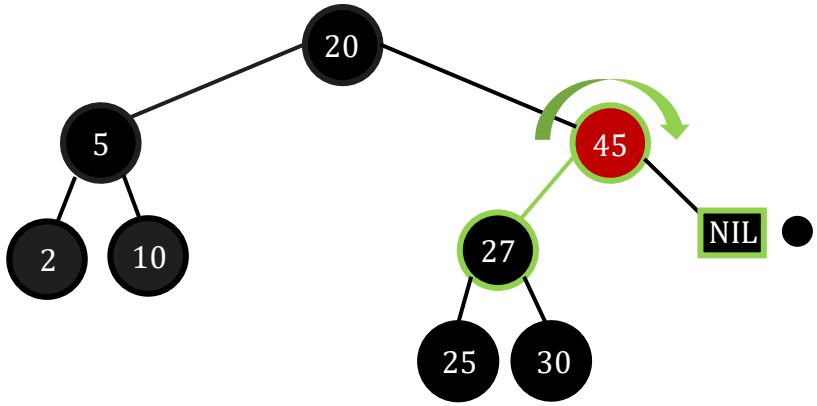
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Examples

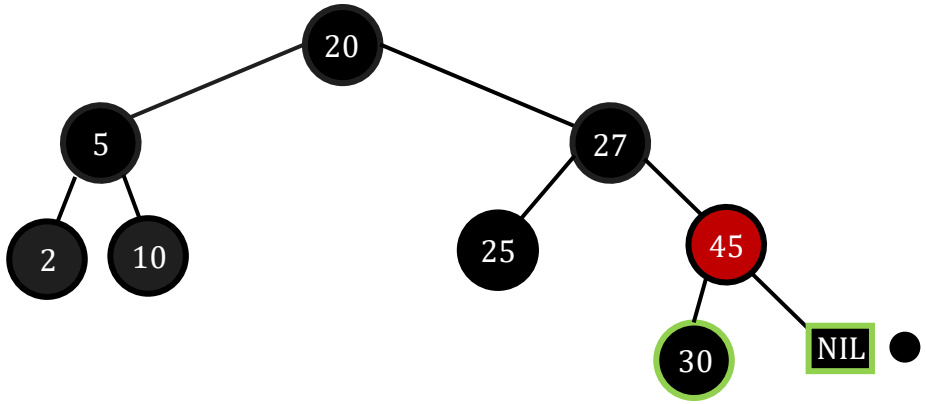
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Examples

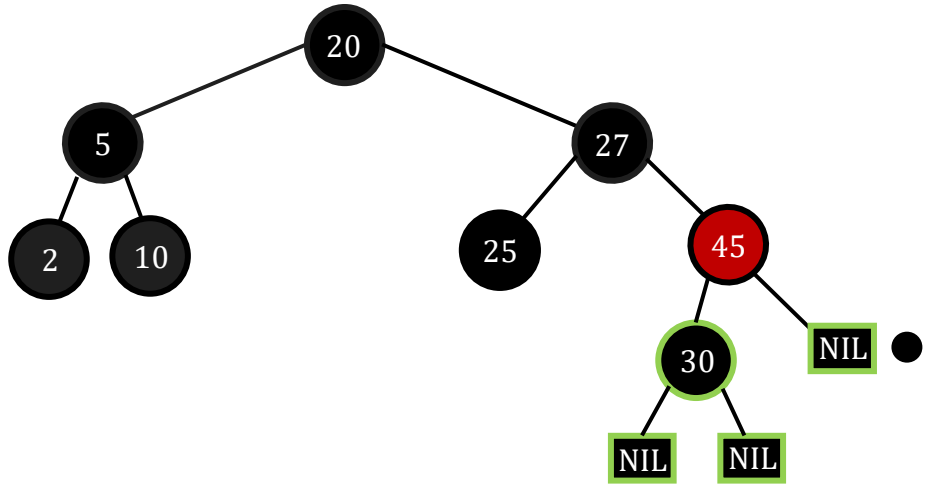
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Examples

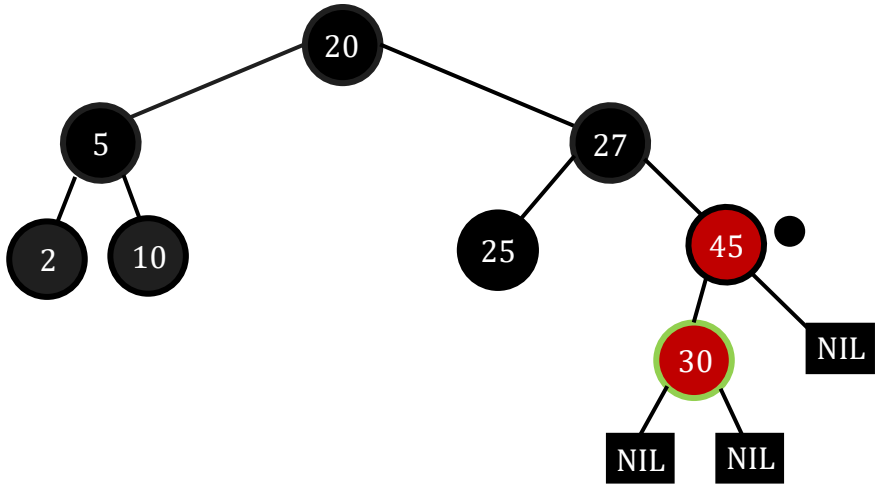
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Examples

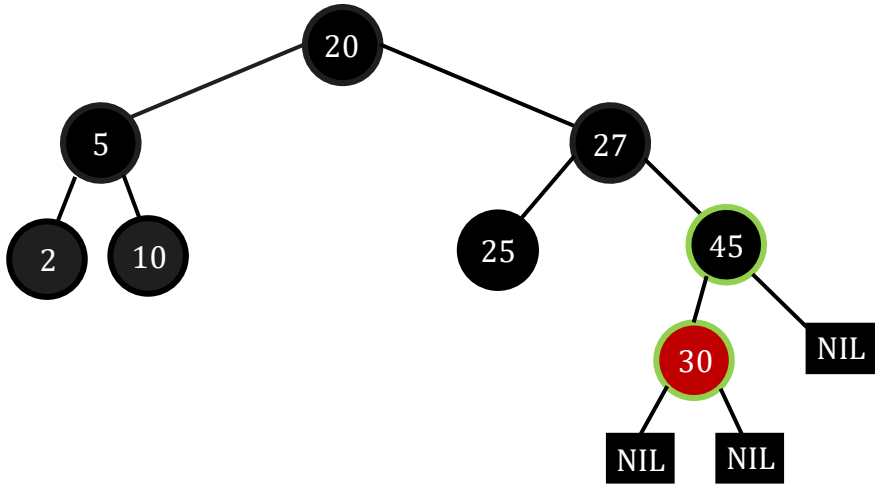
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Examples

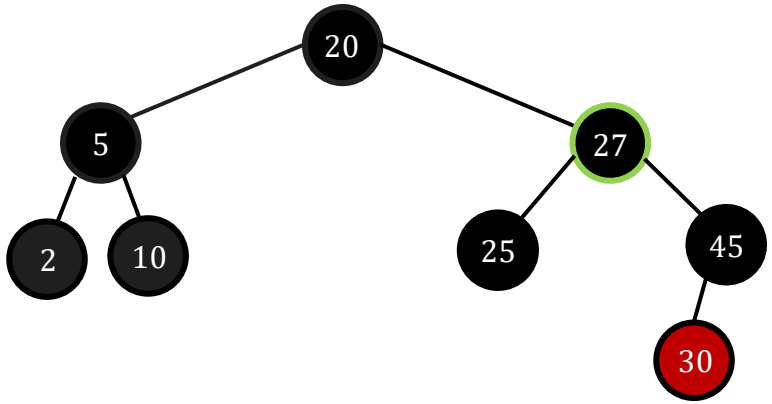
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Examples

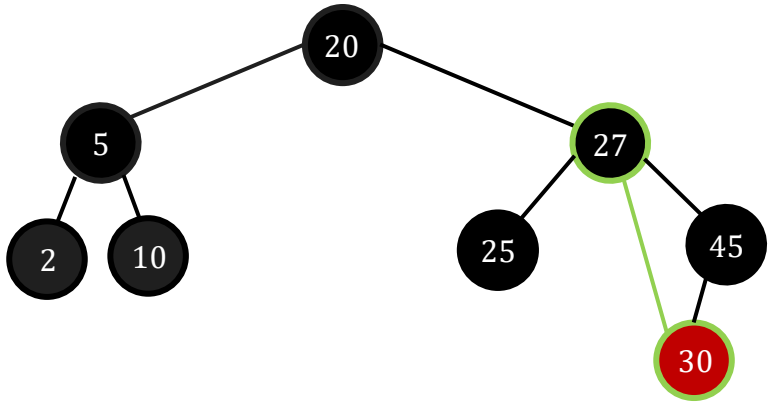
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Examples

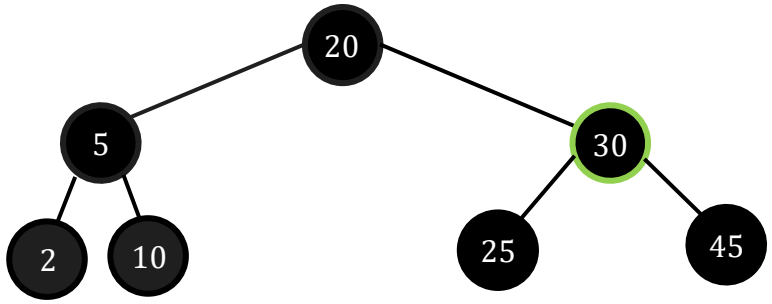
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Examples

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Thank you

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Office : 310-430